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Report No: ICR00003251

IMPLEMENTATION COMPLETION AND RESULTS REPORT

ON

LOAN IDA-5158, LOAN IDA-5514, LOAN IDA-6202 and

TF-15034, TF-15077, TF-A3639, TF B4774

IN THE TOTAL AMOUNT OF

US$ 325,470,000

TO THE

PEOPLE'S REPUBLIC OF BANGLADESH

FOR THE

Rural Electrification and Renewable Energy Development II (RERED II) Project

June 17, 2024

Energy & Extractives Global Practice
South Asia Region

CURRENCY EQUIVALENTS

(Exchange Rate Effective June 17, 2024)

Currency Unit=	BDT
BDT 117.48=	US$1
US$ 1.32=	SDR 1

FISCAL YEAR
July 1 - June 30

Regional Vice President:	Martin Raiser
Country Director:	Abdoulaye Seck
Regional Director:	Pankaj Gupta
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ABBREVIATIONS AND ACRONYMS

BREB	Bangladesh Rural Electrification Board
BCCRF	Bangladesh Climate Change Resilience Fund
CFL	Compact fluorescent lighting
DBD	Domestic biogas digesters
ELIB	Efficient Lighting Initiative of Bangladesh
FAPAD	Foreign Aided Project Audit Directorate
GCF	Green Climate Fund
GPOBA	Global Partnership for Output-Based Aid
ICS	Improved cookstoves
IDCOL	Infrastructure Development Finance Company Limited
IPP	Independent power producer
IRR	Internal rate of return
KfW	Kreditanstalt für Wiederaufbau
kWp	kilowatt peak
LED	Light-emitting diode
MFI	Micro finance institution
MoPEMR	Ministry of Power, Energy and Mineral Resources
MW	megawatt
NGO	Non-Governmental Organization
NDBMP	National Domestic Biogas and Manure Program
PDO	Project development objective
PO	Partner organization
PPP	Public-Private Partnership
PSDTA	Power Sector Development Technical Assistance
RAPSS	Remote Area Power Supply Systems
RERED	Rural Electrification and Renewable Energy Development
SDR	Special Drawing Rights
SHS	Solar home system
SIP	Solar irrigation pump
SMG	Solar mini-grid
SREDA	Sustainable Renewable Energy Development Authority
STEP	Systematic Tracking Exchanges in Procurement
TA	Technical assistance
TF	Trust Fund
USAID	United States Agency for International Development

TABLE OF CONTENTS

DATA SHEET

BASIC INFORMATION

Product Information

Project ID	Project Name
P131263	Rural Electrification and Renewable Energy Development II (RERED II) Project
Country	Financing Instrument
Bangladesh	Investment Project Financing
Original EA Category	Revised EA Category
Partial Assessment (B)	Partial Assessment (B)

Related Projects

Relationship	Project	Approval	Product Line
Additional Financing	P150001-RERED II Additional Financing	19-Jun-2014	IBRD/IDA
Additional Financing	P165400-Additional Financing II for Rural Electrification and Renewable Energy Development II	10-Apr-2018	IBRD/IDA
Supplement	P106135-Grameen Shakti Solar Homes Project	17-Dec-2007	Carbon Offset
Supplement	P107906-Bangladesh - IDCOL Solar Home Systems Project	19-Dec-2007	Carbon Offset
Supplement	P119547-GPOBA: Rural Electrification & Renewable Energy	13-May-2010	Recipient Executed Activities
Supplement	P119549-GPOBA: Bangladesh Solar Home Systems	26-Mar-2010	Recipient Executed Activities

Supplement	P154576-GPOBA Scale-up for Bangladesh RERED II	13-Feb-2015	Recipient Executed Activities

Organizations

Borrower	Implementing Agency
PEOPLE'S REPUBLIC OF BANGLADESH	Infrastructure Development Company Limited (IDCOL), Power Cell

Project Development Objective (PDO)

Original PDO

The proposed project development objectives are to increase access to clean energy in rural areas through renewable energy and promote more efficient energy consumption.

Revised PDO

Increase access to clean energy through renewable energy in rural areas.

FINANCING

		Original Amount (US$)	Revised Amount (US$)	Actual Disbursed (US$)
World Bank Financing				
P131263	IDA-51580	155,000,000	151,171,577	148,307,624
P131263	TF-15034	5,997,500	5,347,536	5,347,536
P131263	TF-15077	10,000,000	9,959,221	9,959,221
P131263	IDA-55140	78,400,000	77,773,964	69,842,040
P131263	TF-A3639	1,072,500	1,072,500	1,072,500
P131263	IDA-62020	55,000,000	29,794,223	24,442,949
P131263	TF-B4774	20,000,000	20,000,000	15,494,584
Total		**325,470,000**	**295,119,021**	**274,466,454**
Non-World Bank Financing				
Borrower/Recipient		42,400,000	0	0
US: Agency for International Development (USAID)		7,600,000	0	0
GERMANY: KREDITANSTALT FUR WIEDERAUFBAU (KFW)		12,900,000	0	0
Local Beneficiaries		53,400,000	0	0
Non-Government Organization (NGO) of Borrowing Country		90,200,000	0	0
Total		**206,500,000**	**0**	**0**
Total Project Cost		**531,970,000**	**295,119,021**	**274,466,455**

KEY DATES

Project	Approval	Effectiveness	MTR Review	Original Closing	Actual Closing
P131263	20-Sep-2012	20-Feb-2013	06-Sep-2015	31-Dec-2018	18-Dec-2023

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Amount Disbursed (US$M)	Key Revisions
28-Jan-2014	62.97	Change in Loan Closing Date(s) Reallocation between Disbursement Categories
28-May-2015	132.80	Reallocation between Disbursement Categories
16-Jun-2016	181.95	Change in Loan Closing Date(s)
25-Aug-2016	189.61	Change in Results Framework Change in Loan Closing Date(s) Change in Disbursements Arrangements
16-Jun-2017	212.28	Change in Results Framework Change in Components and Cost Change in Loan Closing Date(s) Reallocation between Disbursement Categories
21-May-2021	245.37	Change in Results Framework Change in Loan Closing Date(s) Reallocation between Disbursement Categories Change in Implementation Schedule
12-Dec-2023	271.18	Cancellation of Financing Reallocation between Disbursement Categories

KEY RATINGS

Outcome	Bank Performance	M&E Quality
Satisfactory	Moderately Satisfactory	Modest

RATINGS OF PROJECT PERFORMANCE IN ISRs

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US$M)
01	01-Jan-2013	Satisfactory	Satisfactory	0
02	20-Jun-2013	Satisfactory	Satisfactory	10.27
25	18-Jan-2014	Moderately Satisfactory	Moderately Satisfactory	62.72
26	28-Jun-2014	Moderately Satisfactory	Moderately Satisfactory	91.34
27	15-Dec-2014	Satisfactory	Moderately Satisfactory	118.07
28	10-Jun-2015	Moderately Satisfactory	Moderately Satisfactory	132.95

29	30-Dec-2015	Moderately Satisfactory	Moderately Satisfactory	164.61
30	09-Jun-2016	Moderately Satisfactory	Moderately Satisfactory	181.95
31	18-Aug-2016	Moderately Satisfactory	Moderately Satisfactory	189.10
32	27-Feb-2017	Satisfactory	Satisfactory	202.55
33	08-Sep-2017	Satisfactory	Satisfactory	213.32
34	16-Mar-2018	Satisfactory	Satisfactory	215.39
35	20-Oct-2018	Satisfactory	Satisfactory	218.59
36	10-Apr-2019	Satisfactory	Satisfactory	225.36
37	22-Oct-2019	Satisfactory	Satisfactory	230.55
38	08-Jun-2020	Satisfactory	Satisfactory	235.75
39	23-Dec-2020	Satisfactory	Satisfactory	242.60
40	20-Sep-2021	Satisfactory	Moderately Satisfactory	245.69
41	09-Apr-2022	Moderately Satisfactory	Moderately Satisfactory	252.22
42	22-Oct-2022	Moderately Satisfactory	Moderately Satisfactory	255.14
43	27-Apr-2023	Moderately Satisfactory	Moderately Satisfactory	266.01

SECTORS AND THEMES

Sectors

Major Sector/Sector	(%)
Energy and Extractives	**100**
Public Administration - Energy and Extractives	3
Renewable Energy Solar	85
Other Energy and Extractives	12

Themes

Major Theme/ Theme (Level 2)/ Theme (Level 3)	(%)

Private Sector Development	**18**
Business Enabling Environment	8
Investment and Business Climate	8
Public Private Partnerships	10
Social Development and Protection	**3**
Social Inclusion	3
Participation and Civic Engagement	3
Human Development and Gender	**2**
Gender	2
Urban and Rural Development	**45**
Rural Development	45
Rural Infrastructure and service delivery	45
Environment and Natural Resource Management	**42**
Climate change	42
Mitigation	42

ADM STAFF

Role	At Approval	At ICR
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I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

Context

Country - **In 2012, Bangladesh benefited from strong and sustained GDP growth, although this was jeopardized by suboptimal energy supply.** The country sustained a high growth trajectory, with an average GDP growth above 6 percent in recent years. Poverty was halved by 2015, meeting the Millennium Development Goal. However, infrastructure deficits remained high, particularly in the energy sector. Lack of access to power posed the main threat to a sustained GDP and exports growth; for instance, 70 percent of firms reported experiencing electrical outages[1].

Sector **- At the time of RERED II appraisal, a large segment of the population of Bangladesh had little or no access to *electricity***. Only 55 percent of the population[2] had access to electricity, a rate that drops to 43 percent in rural areas. Per capita electricity consumption was about 236 kilowatt-hours (kWh) per year, one of the lowest in the world. Due to the high cost of thermal power plants, the volatility of fossil fuel prices, and unsustainable subsidy policies to the electricity and petroleum sectors, off-grid renewable energy solutions seemed the only near-to-medium term option for remote areas. However, scaling-up the development of renewable energy was hampered by the absence of an overarching institution dedicated to renewable energy policy design in Bangladesh.

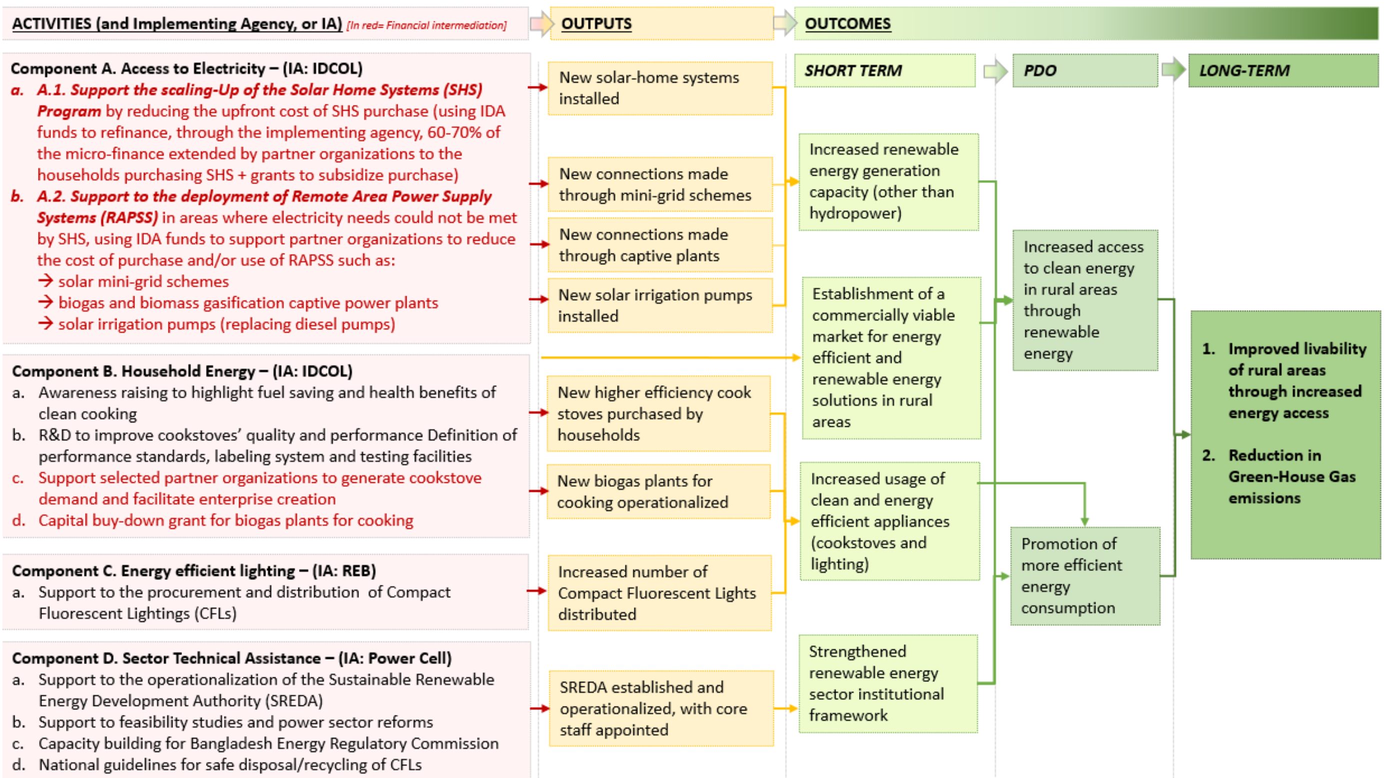
Similarly, a large segment of the population of Bangladesh had little or no access to *efficient and/or clean energy sources*. Out of the total 30 million households in Bangladesh, 90 percent used traditional biomass fuels like fuelwood for cooking in low-efficiency stoves[3]. Incomplete combustion produced severe indoor air pollution[4]. In addition, most households used highly energy-inefficient incandescent lighting, further burdening modest rural households with higher electricity bills. Clean cooking and energy-efficient lighting solutions were not broadly adopted due to lack of awareness, poor quality of cooking supplies, and higher initial cost compared to traditional biomass fuels.

Rationale for Bank Support **- To address these challenges, the design of the RERED II built on and scaled up activities supported by the RERED I (P071794[5]).** The project development objective (PDO) of the RERED I was to increase access to electricity in rural areas of Bangladesh and help promote efficient energy consumption. The Global Environment Objective of the RERED I was to reduce atmospheric carbon emissions by overcoming market barriers for renewable energy development, including high implementation costs. At closing, RERED I had achieved all its PDO-level targets. The RERED II was therefore conceived to continued supporting the Government to increase access to clean energy in rural areas through renewable energy and more efficient energy consumption, while (i) using the Government existing initiatives, and (ii) building on the achievements and lessons learned of the RERED I.

[1] World Bank, Enterprise Surveys – 2013
[2] Approximately 150 million in 2012
[3] 5 to 15 percent fuel efficiency
[4] Pollution level exceeding 200µg/m3, well above the daily standard of 25µg/m3 established by the World Health Organization
[5] Approved in 2002 and closed in December 2012

Theory of Change (Results Chain)

The Project Appraisal Document of the RERED II Project did not explicitly describe a theory of change. However, a chain can be derived from the project description and results framework.



Key underlying assumptions: (i) Grid expansion in the project areas would only take place in the medium to long term; (ii) Customers (including low-income households) have a willingness-to-pay for improved energy services, (iii) selected partner organizations (POs) have sufficient local presence and benefit from final beneficiaries' trust, making them key intermediaries in the deployment of the suggested solutions.

Figure 1. Implicit Theory of Change at appraisal

The long-term outcomes envisioned at appraisal were to improve the livability of rural areas and reduce greenhouse gas (GHG) emissions. These outcomes were expected to result from a holistic approach covering several key pillars of the energy sector (see Figure 1 and components below).

This vision was implemented through an innovative Public-Private Partnership (PPP) to create a market-driven renewable energy sector in rural areas. This PPP relied on the association of: (i) the main government-owned non-bank financial enterprise operating in infrastructure finance in Bangladesh (Infrastructure Development Company Limited, or IDCOL); and (ii) a comprehensive network of private actors such as microfinance institutions (MFIs), sponsors and developers, manufacturers, non-governmental organizations (NGOs) and other entities (laboratories, testing facilities, etc.). Power Cell[6] would be in charge of the institutional strengthening aspect.

[6] Government regulatory agency under the Power Division of the Ministry of Power, Energy and Mineral Resources

Project Development Objectives (PDOs)

At appraisal and as agreed upon in the financing agreement[7], the objective of the Project was "to increase access to clean energy in rural areas through renewable energy and to promote more efficient energy consumption".

Key Expected Outcomes and Outcome Indicators

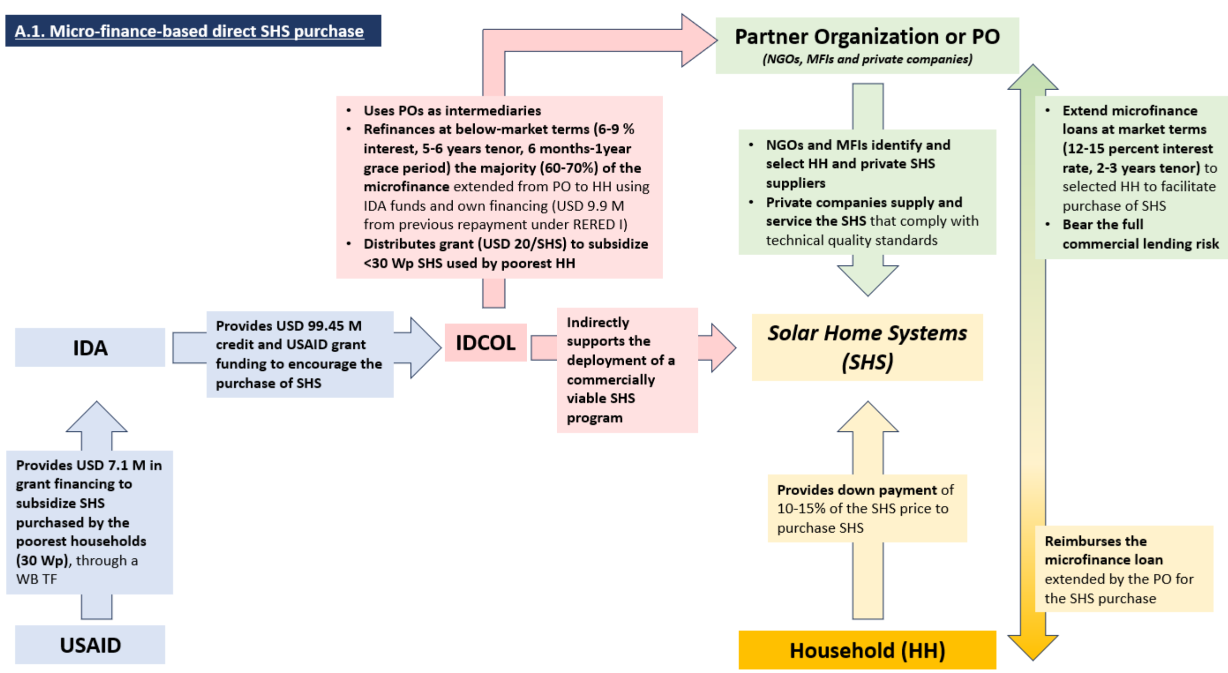
At appraisal, the Project had two expected outcomes derived from the PDO: (i) increase access to clean energy in rural areas through renewable energy, and (ii) promote efficient energy consumption. The outcome indicators (PDO-level) described in the original Project Appraisal Document (PAD) were:

(i) Households, farmers, and businesses having access to clean energy services (Number);
(ii) Generation capacity of Renewable Energy (other than hydropower) constructed (MW);
(iii) Efficient energy consumption through introduction of energy efficient lighting (MW saved per year);
(iv) Direct project beneficiaries, including female beneficiaries (Number)

Components

0. **The project was designed to have four independent components, acting as four different sub-projects under the same umbrella project**. For reference, photographs of disseminated solutions are provided in Annex 6.2.

1. **Component A. Access to Electricity (Appraisal financing estimate – Total: US$ 309.2 million; IDA: US$ 116 million)** ***Sub-component A.1. Solar Home Systems (SHS) (Appraisal financing estimate - Total: US$ 199.8 million; IDA: US$ 99.45 million) –*** Following RERED I implementation arrangements, this component used IDA funding to encourage the purchase of SHS through a micro-finance scheme to improve the affordability of the suggested solutions (*see figure 2*). In parallel, the government also supported the SHS market through fiscal expenditure, including the exemption of duty imports for solar panels and raw materials used in the manufacturing of batteries. The scheme is:



[7] Financing Agreement signed on October 23, 2012, for the IDA credit 5158-BD.

Figure 2. Description of Component A.1. activities – Solar Home Systems microfinance-based sale

Sub-component A.2. Remote Area Power Supply Systems (RAPSS) (Appraisal financing estimate – Total: US$ 109.4 million, IDA: US$ 16.55 million) – This component was designed to meet the electricity needs of rural households and businesses that could not be met with SHS or on-grid solutions in the short to medium term. Several clean energy solutions were proposed to be implemented under the RAPSS guidelines, with different modus operandi, as follows:

➔ ***A.2.i. Mini-grid and bioenergy projects –*** This sub-activity was designed to use IDA funding to alleviate barriers faced by private sponsors of RAPSS projects, to encourage private-sector-led grid-quality renewable energy generation projects. Such barriers included insufficient financial support, lack of technical assistance, and poor-quality control mechanisms. This sub-activity encouraged at least two types of off-grid technologies: (i) solar photovoltaic (PV) mini-grids, and (ii) biomass-based captive plants. The scheme is as follows (*See Figure 3*):

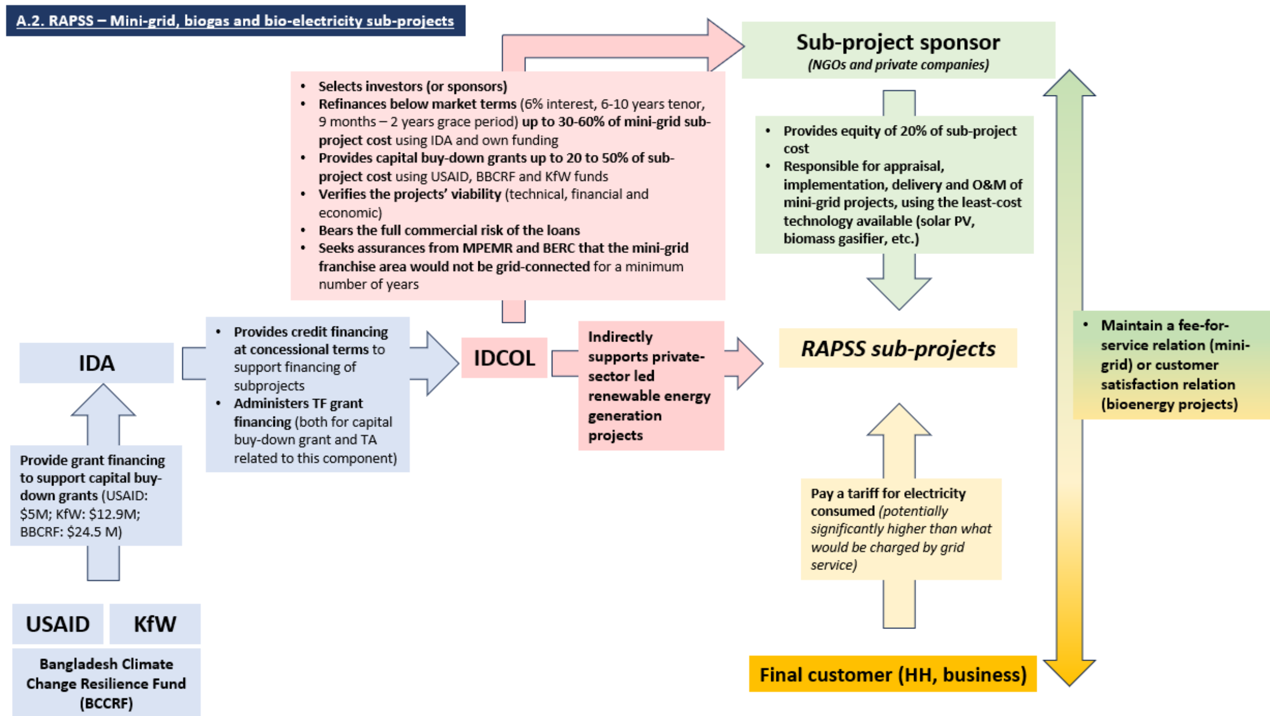


Figure 3. Description of Component A.2.i. activities – RAPSS sub-projects schemes

➔ ***A.2.ii. Solar-powered irrigation pumps –*** This sub-activity was designed to use IDA funding to address the challenges faced by farmers using diesel-powered or electric irrigation pumps requiring significant amounts of subsidized electricity[8]. Challenges included: (i) heavy fiscal and household out-of-pocket pump cost; (ii) hindered agricultural productivity and profitability (cost of energy input and maintenance of diesel pumps); and (iii) environmental and health adverse effects due to diesel combustion. This sub-activity used IDA funding to encourage the purchase of water from solar-powered irrigation pumps (see Figure 4).

[8] At the time of appraisal, 0.34 million electric pumps were in use – Source: IDCOL.

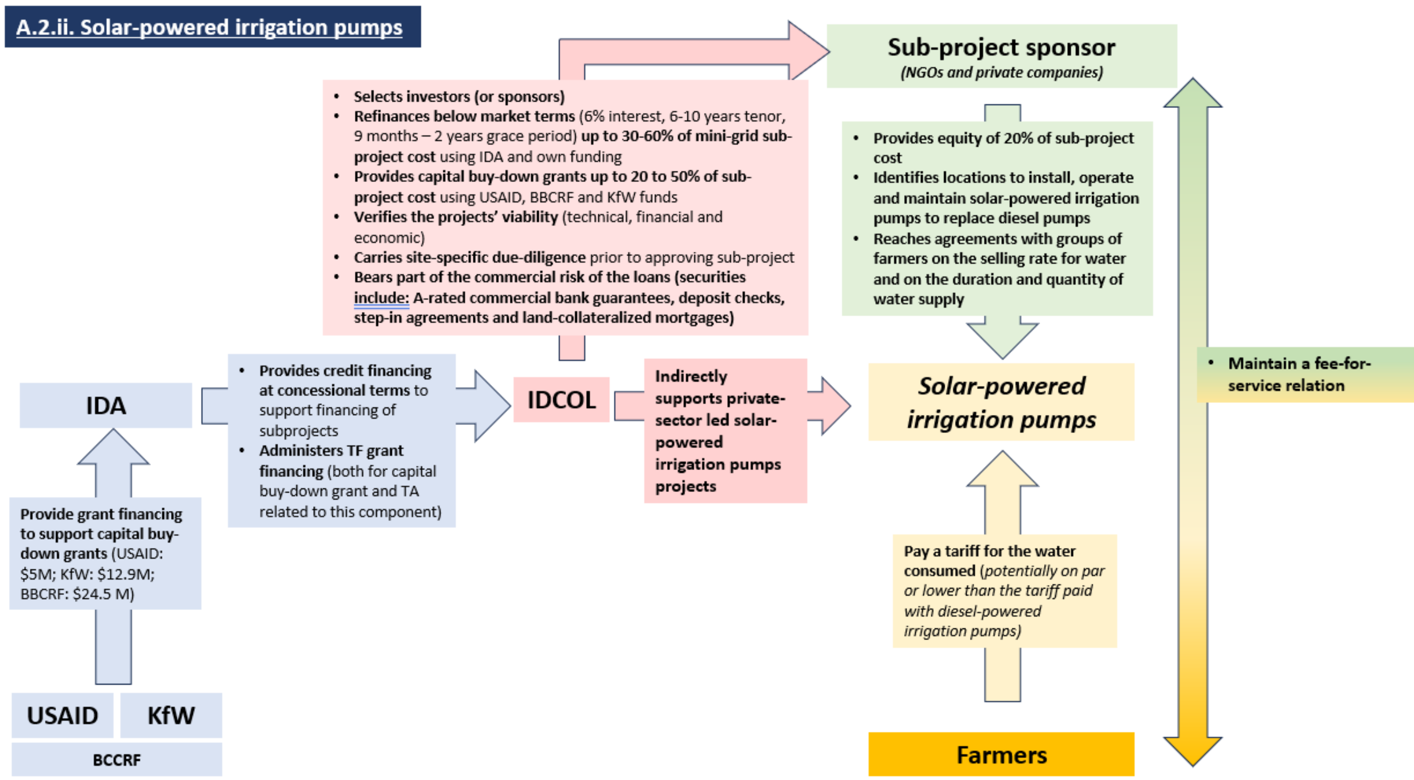
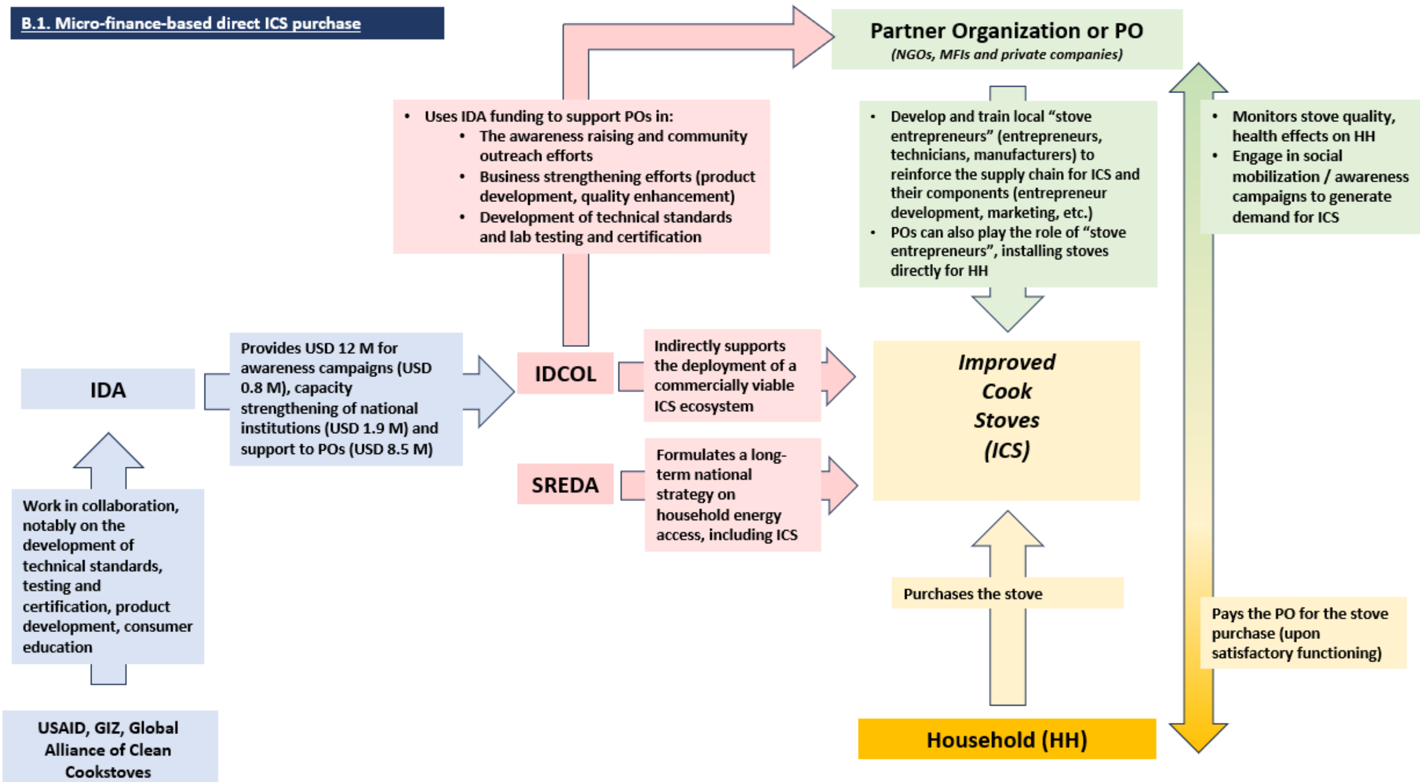


Figure 4. Description of Component A.2.ii. activities – Solar-powered irrigation subprojects for fee-for service scheme

2. **Component B. Household Energy (Appraisal financing estimate – Total: US$ 46.3 million, IDA: US$ 12 million)** – This component was designed to meet the cooking energy needs of households in rural areas where biomass combustion remained the predominant form of energy used, raising health and safety risks. The main factors hindering the adoption of modern cooking solutions were: (i) lack of widespread awareness campaigns; (ii) affordability challenges; (iii) absence of a sustainable funding mechanism for scaled programs; (iv) cost-revenue shortfalls for enterprises in the sector, preventing the development of a strong private sector promoting cooking solutions; and (iv) an overall lack of oversight on the sector, hindering the development of strong technical standards and best practices. To address these challenges, two clean cooking energy solutions were supported under this component: improved cook stoves (ICS) and domestic biogas digesters (DBD), assuming other modern technologies like natural gas, liquefied petroleum gas (LPG) or electricity would not be easily accessible to the targeted population in the near term. ICS and DBD were deployed through different modus operandi, as follows:

- ***B.1. Improved Cook Stoves (ICS) deployment –*** Demand creation and supply chain development to encourage direct ICS purchase from households, based on the following arrangements (*Figure 5)*:



NB: SREDA = Sustainable Renewable Energy Development Authority

Figure 5. Description of Component B.1. activities – Household Energy | ICS purchase

- ***B.2. Domestic biogas digesters (or DBD) with biogas stoves for cooking energy –*** Demand creation and capital buy-down for the construction of the biogas plants to encourage biogas use from households, based on the implementation arrangement of an existing program[9] (see Figure 6):

[9] Biogas program funded by the Netherlands Development Organization (SNV), scaled up by IDA under the RERED II Project.

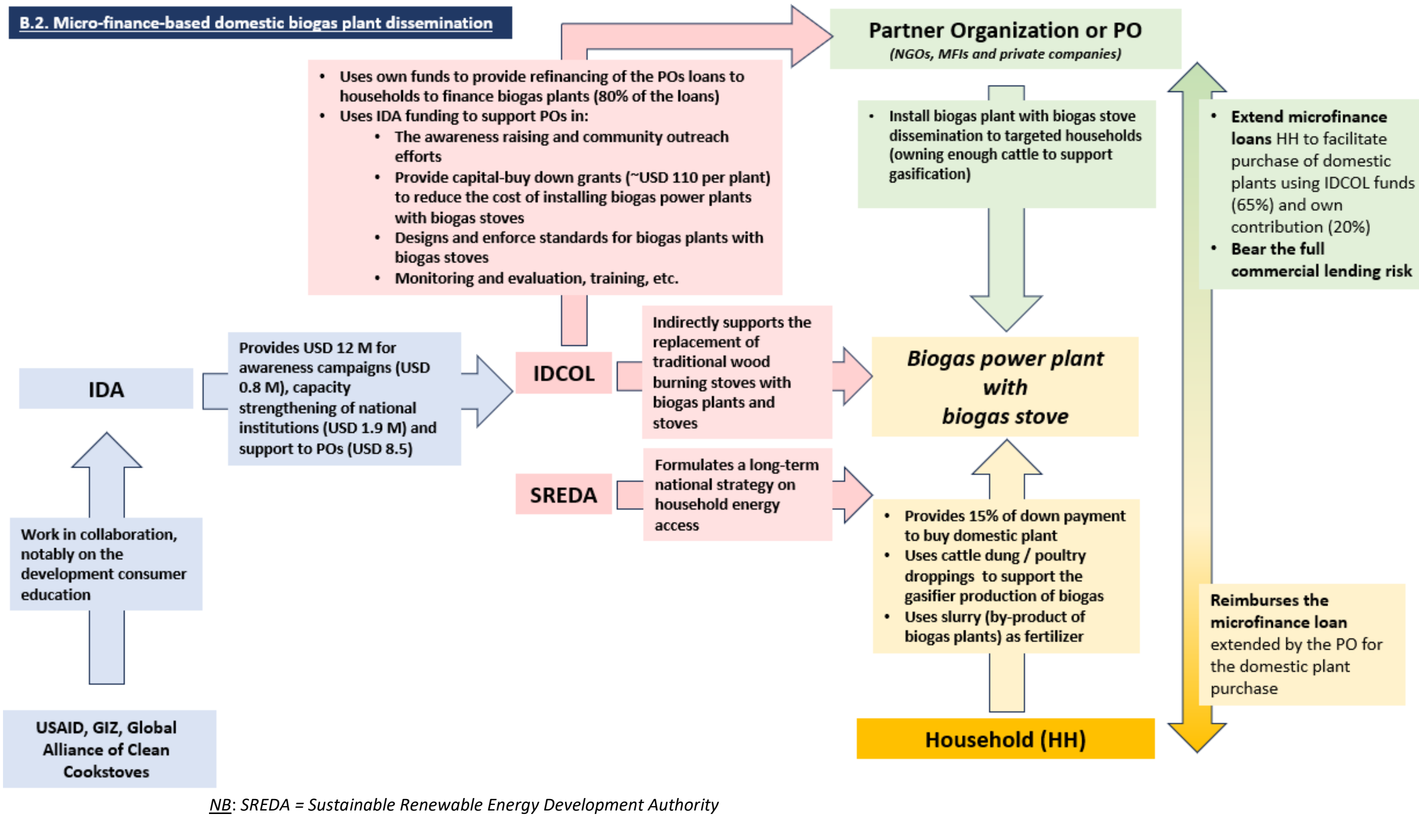


Figure 6. Description of Component B.2. activities – Household Energy | Domestic biogas plant purchase

Component C. Energy efficient lighting (Appraisal financing estimate – Total: US$ 19 million, IDA: US$ 17 million) – This component was designed to reduce the peak energy demand in Bangladesh by reducing the electricity consumption of highly inefficient incandescent light bulbs, used by 28 million households according to a 2008 GIZ study. To address this challenge, this component aimed at implementing the second phase of an existing program designed for this purpose: the Efficient Lighting Initiatives of Bangladesh (ELIB). This program, launched in 2009, aimed to distribute compact fluorescent lamps (CFLs) for free in exchange for incandescent lamps used by households. This program would be deployed in two phases: 10.5 million CFLs in phase 1, and 17.5 million CFLs in phase 2. While phase 1 was completed before the appraisal of RERED II, customer feedback pointed out serious quality concerns regarding the CFLs distributed. In addition, the preparation of phase 2 raised serious concerns about procurement fraud at the bidding stage. Therefore, to encourage successful implementation of phase 2 of ELIB, this component was introduced under the RERED II scope, designed to learn from ELIB's previous challenges (for example, revision of the bidding documents with the help of an international technical consultant to strengthen clauses for quality assurances, pre-shipment and post-shipment inspection and testing). This component was implemented based on the following implementation arrangement (*See Figure 7):*

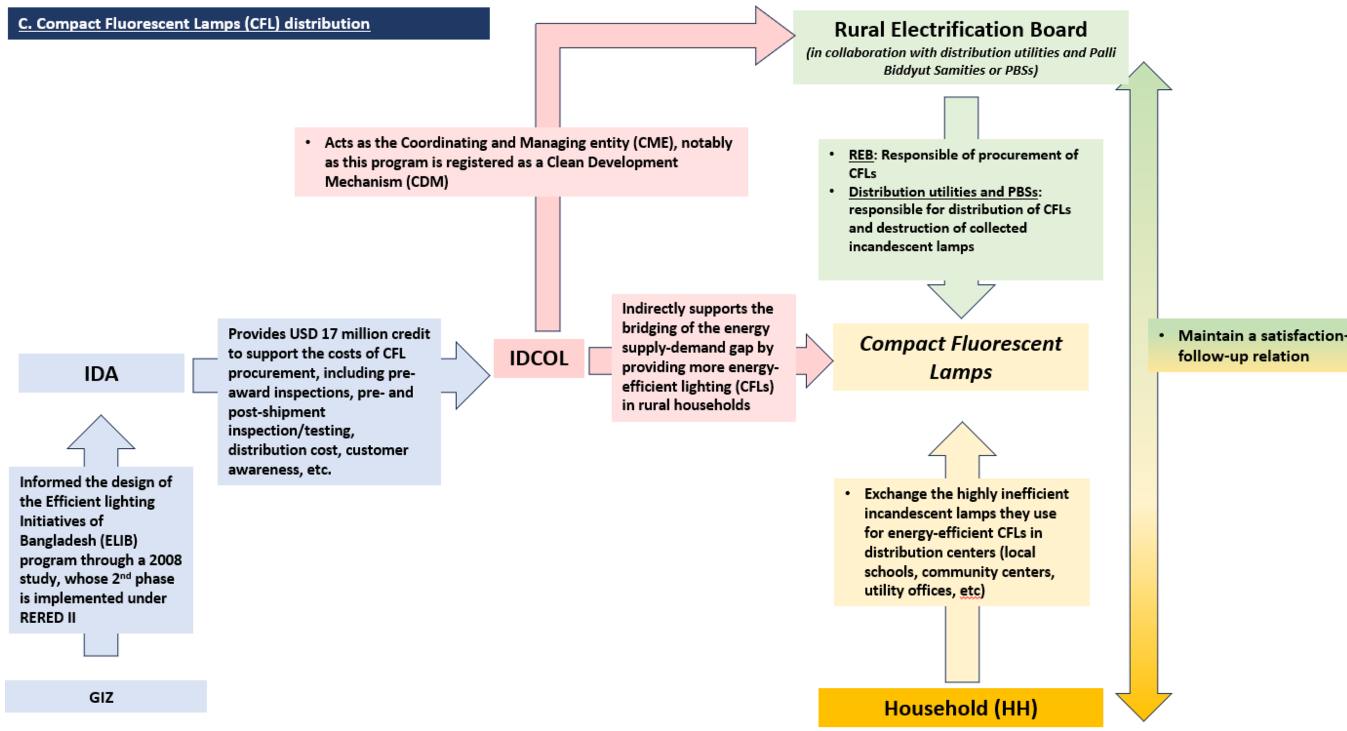


Figure 7. Description of Component C. activities – Efficient Lighting | CFL Distribution

3. **Component D. Sector Technical Assistance (Appraisal financing estimate – Total: US$ 6.5 million, IDA: US$ 5 million) –** This component was designed to provide technical assistance to various energy sector institutions involved in the promotion and development of renewable energy in Bangladesh. The activities implemented under this component are a continuation of an existing Bank-financed project that closed in 2012: the Power Sector Development Technical Assistance (PSDTA) project. In continuation of the PSDTA activities beyond closing, this sub-component was designed to be implemented by Power Cell (the technical arm of the Power Division of the Ministry of Power, Energy and Mineral Resources – MoPEMR) to carry out the following:
    - Support for the establishment of the Sustainable and Renewable Energy Development Authority (SREDA)
    - Support for the Bangladesh Energy Regulatory Commission (BERC) for capacity building
    - Activities related to power sector reform roadmap, project feasibility and preparation studies
    - Training, road shows, workshops, seminars, and study tours for power sector capacity building
    - Selective monitoring, evaluation, and coordination activities related to RERED II
    - Office equipment support and incremental operating costs for Power Cell.

4. ***Implementation arrangements* – While forming part of an overarching umbrella Project (RERED II), each component and sub-component was designed to be implemented independently**: IDCOL was responsible for overall implementation of Component A and B through one Project Management Unit (PMU) dedicated to each component, while REB was responsible of the implementation of Component C, and Power Cell for Component D. Under Component A and B and their respective PMUs, IDCOL appointed dedicated project supervisors to each sub-component, allowing each sub-activity to be monitored as a stand-alone activity.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION (IF APPLICABLE)

Revised PDOs and Outcome Targets

5. ***PDO change*** – **The PDO was changed** from "*Increase access to clean energy in rural areas through renewable energy and promote more efficient energy consumption*" to "*Increase access to clean energy through renewable energy in rural areas*". This change took place during the first RERED II restructuring in June 2014 (*see section on 'Key Changes')*. The change was made to reflect the deletion of Component C 'Efficient Lighting'. While the remaining activities also indirectly contributed to energy efficiency, the team wished to monitor progress mostly on energy access, believing it would not have been incentivized otherwise. This rationale can be challenged (see 'Efficacy').

6. ***Outcome targets*** – **Two main major revisions of targets can be observed throughout project implementation**. An initial major revision of outcome targets occurred in 2014 (first Level I restructuring). As the component on efficient lighting was removed, its related targets were dropped. In addition, different sub-project dynamics (fast SHS uptake, slow ICS deployment) led to various outcome target revisions (see Table 1). A second major revision of outcome targets occurred in June 2018 (second Level I restructuring), as project activities significantly picked up in their pace of deployment, aided by changes in implementation modalities (see section on 'Factors subject to Implementing Agencies' control') and the results of Additional Financing II mobilization (See 'Other Changes'). The results framework was also improved by deleting redundant indicators (for instance, 'generation capacity of renewable energy constructed solar', and 'people that gain access only through switching cooking and/or heating systems').

Revised PDO Indicators

7. **Over the course of implementation, the PDO-level indicators were changed both in nature and in targets.** *In nature*, the PDO-level indicators were modified to reflect the changes in the contour of the project. For instance, the PDO-level indicator 'Efficient energy consumption through introduction of energy efficient lighting (MW saved per year)' was dropped. New indicators to monitor increased energy access were added (see Table 1 below). *In targets*, the PDO-level and intermediate indicators were modified to reflect project progress and challenges (See Table 1 below).

8. **While the changes described in section B 'Significant changes' are highly relevant to address observed project dynamics, this ICR discusses the missed opportunities related to these restructurings.** Missed opportunities include the opportunity to further orient the results-framework to a more-outcome oriented approach, or measure key outcomes such as market development or environmental benefits (see section 'M&E Design'). In addition, the number of restructurings, their timing, and their nature (two level I restructurings, including only two years after project effectiveness) is both a sign of great flexibility enabled by the project implementation arrangement, but also signals a potential missed opportunity for a better framing during project preparation (see section 'Quality at entry' and 'Lessons Learned').

Indicator Name	Unit	Target	*Revised targets **					
PDO indicators		***As per PAD***	**Jan 2014**	**June 2014**	**Aug 2016**	**Jun 2018**	**May 2021**	**December 2023**
Number of households, farmers, and businesses having access to clean energy services	Number	1,578,000	1,578,000	2,069,000	2,069,000	6,085,082	6,085,082	6,085,082
Generation Capacity of Renewable Energy (other than hydropower) constructed	MW	61	61	56	56	88	88	88
More efficient energy consumption through introduction of energy-efficient lighting	MW saved per year	160	160	***DROPPED***	*N/A*	*N/A*	*N/A*	*N/A*
Generation Capacity of Renewable Energy constructed-Solar	MW	*N/A*	*N/A*	***NEW*** - 56	56	***DROPPED***	*N/A*	*N/A*
Direct project beneficiaries	Millions	10.4	10.4	5.7	5.7	9.8	9.8	9.8
Female Beneficiaries	Percentage	55	55	59	59	76	76	76
Projected lifetime energy savings	Megawatt hour (MWh)	*N/A*	***NEW -*** 480,000	***DROPPED***	*N/A*	*N/A*	*N/A*	*N/A*
People provided with access to electricity under the project by household connections	Millions	*N/A*	***NEW*** – 2.5	4.7	4.7	5.5	5.5	5.5
People who gained access to more energy-efficient cooking and/or heating facilities	Millions	*N/A*	***NEW*** – 4.3	1	1	5	5	5
People that gained access only through switching cooking and/or heating systems	Millions	*N/A*	***NEW*** – 4.3	1	1	***DROPPED***	*N/A*	*N/A*

Indicator Name	Unit	Target	*Revised targets**					
Intermediate Results indicators		***As per PAD***	**Jan 2014**	**June 2014**	**Aug 2016**	**Jun 2018**	**May 2021**	**December 2023**
Component A – Access to electricity								
Number of solar home systems installed	Number	550,000	550,000	1,030,000	1,030,000	1,202,174	1,202,174	1,202,174
Number of connections made through mini-grid systems and captive plants	Number	6,750	6,750	7,500	7,500	28,100	26,881	26,881
Collection efficiency of the SHS POs	Percentage	*N/A*	*N/A*	***NEW*** – 90%	90	90	90	90
Number of solar irrigation pumps installed	Number	1,500	1,500	1,250	1,250	2,170	1,987	1,987
Component B – Household Energy								
Number of improved cook stoves purchased by households	Number	1,000,000	1,000,000	1,000,000	1,000,000	5,000,000	5,000,000	5,000,000
Number of biogas plants installed	Number	20,000	20,000	33,000	16,000	8,300	8,300	8,300
Component C – Efficient Lighting (Dropped)								
Number of energy efficient lamps distributed	Number	7,250,000	7,250,000	***DROPPED***	*N/A*	*N/A*	*N/A*	*N/A*
Component D – Sector Technical Assistance (later called "Component C")								
Enabling policy for renewable energy development	Text	SREDA Operational with core staff appointed	SREDA Operational with core staff appointed	SREDA Operational with core staff appointed	SREDA Operational with core staff appointed	SREDA Operational with core staff appointed	SREDA Operational with core staff appointed	SREDA Operational with core staff appointed
Number of female staff in POs	Number	*N/A*	*N/A*	*N/A*	*N/A*	*N/A*	***NEW*** - 480	480
Grievances received that are addressed within two month of receipt	Percentage	*N/A*	*N/A*	*N/A*	*N/A*	*N/A*	***NEW*** - 100	100

**Green cells reflect an increase in target; Red cells reflect a decrease in target; Grey cells reflect an inexistent target (either not included at appraisal or dropped)*

Table 1: Summary of changes in PDO-level and intermediate indicator targets

Revised Components

9. Component C. 'Energy Efficient Lighting' was dropped during a June 2014 restructuring (see below), as evidence suggested that CFL uptake was already in effect without concessional financing or free distribution needed. Its funding was reallocated to Component A. 'Access to Electricity' and Component D. 'Sector Technical Assistance'.

Other Changes

0. Over the course of implementation, at the request of the Economic Relations Division (ERD), Ministry of Finance (MoF), the Project underwent seven restructurings. This section focuses on the most significant ones (Level I restructurings and changes in closing dates). Annex 7 provides additional information.

(i) **In June 2014, a Level I restructuring** aimed at (i) mobilizing an additional IDA credit (Additional Financing I or AF I – IDA credit 5514-BD) of SDR 50,585,012 (US$ 78,400,000) and (ii) dropping the 'Energy Efficient Lighting' Component. This led to a change in PDO, results framework and financing allocation between components.

(ii) **In June 2018, a second Level I restructuring** allowed the project to mobilize a second additional IDA credit (AF II – IDA credit 6202) of SDR 37,800,000. This restructuring resulted in an upward revision of nearly all outcome targets (see Table 1). This was also the opportunity to extend the closing date from December 31, 2018, to December 31, 2021, to provide ample time for these new ambitions to reach fruition.

(iii) **In May 2021**, the Project was restructured to (i) extend the closing date from December 31, 2021, to December 18, 2023; (ii) reallocate US$ 0.45 million between categories within Component A; and (iii) revise intermediary indicators and end target values to reflect the extension.

While all restructurings do not appear explicitly in the Portal-generated list of restructurings due to Project supervision shortcomings (see 'Bank Performance'), they have been reconstituted and inferred from project and legal documents.

1. Furthermore, the RERED II project benefited from additional grants from various development partners, such as:

Donor	TF #	Amount	Approval	Closing	Component
Administered through the World Bank system – Recipient-executed (RE)					
GPOBA Mini-Grid	TF096552	US$ 1.1 million	May 12, 2010	Jun 30, 2015	A. RAPSS (SMG)
USAID	TF015034	US$ 5.34 million	Jun 24, 2013	Dec 31, 2018	A. SHS, RAPSS
BCCRF	TF015077	US$ 9.96 million	Jul 09, 2013	Dec 31, 2016	A. RAPSS (SIP)
GPOBA	TF019156 TF019157	US$ 15 million	April 23, 2015	Jun 30, 2017	B. HH Energy (DBD)
USAID	TF072675	US$ 1.07 million	Oct 06, 2016	Dec 31, 2018	A. SHS
Green Climate Fund (GCF)	TF0B4774	US$ 20 million	Dec 16, 2020	Dec 18, 2023	B. HH Energy
Administered through the World Bank system – Bank-executed (BE)					
ESMAP	TF011458	US$ 140,022	Dec 21, 2011	Dec 31, 2015	B. HH Energy
ESMAP	TF0A8133	US$ 393,211	Jul 17, 2018	Nov 30, 2022	B. HH Energy
Non-exhaustive parallel co-financing (excluding co-financing from the government, the POs, and financial beneficiaries)					
USAID	N/A	US$ 7.6 million	2012	-	A. SHS, RAPSS
KfW	N/A	US$ 12.9 million	2013	-	A. RAPSS

Table 2: Grant financing introduced throughout the implementation of the project

Rationale for Changes and Their Implication on the Original Theory of Change

22. The deletion of Component C 'Energy Efficient Lighting' encouraged the team to narrow the results framework (and inferred theory of change) to 'increased access to clean energy in rural areas through renewable energy', such as:

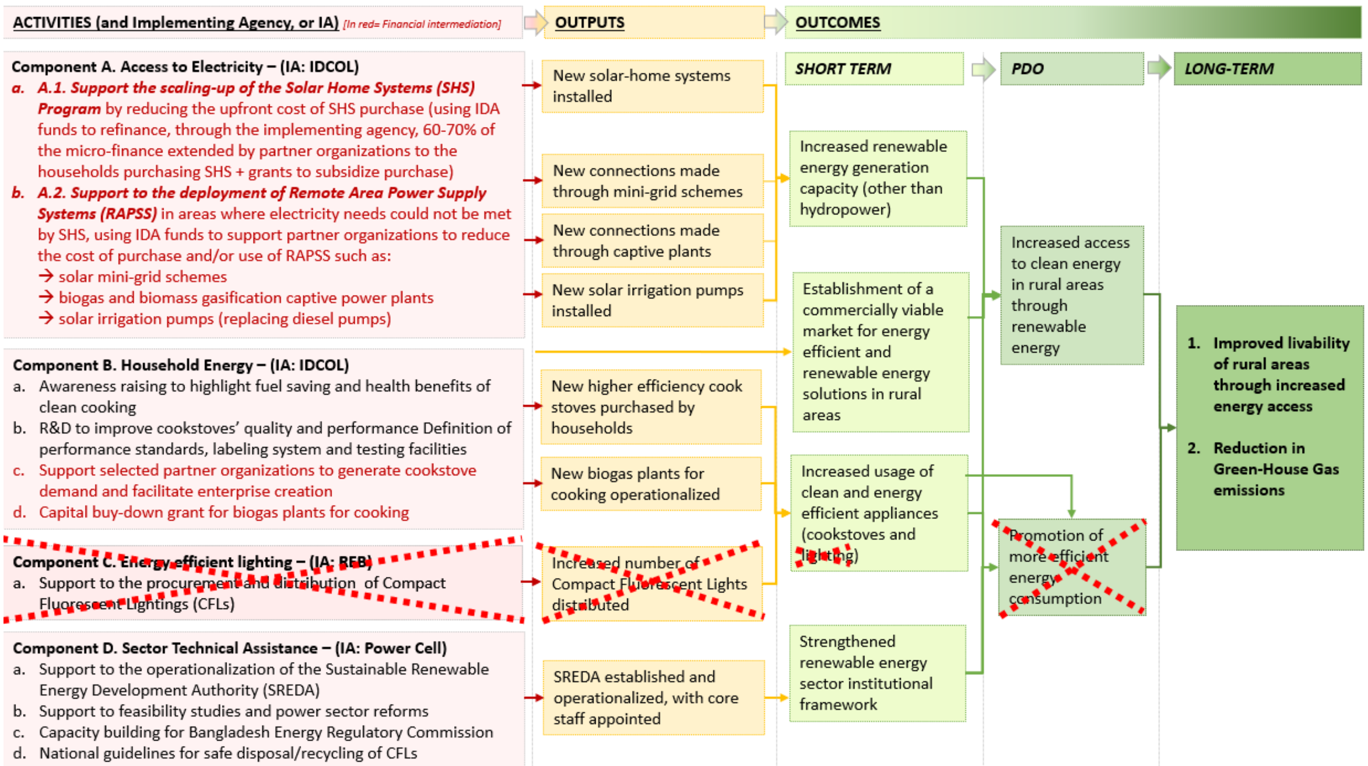


Figure. 8. Implications of project design change on the original Theory of Change

23. The rationale for the presented changes were the following (See Annex 7 for other changes):

i. **June 2014:** The government of Bangladesh requested additional financing I (AF I – IDA Credit 5514) from the World Bank to meet the funding needs of the fast-growing SHS program (Component A.1). Component C was no longer deemed relevant, as an uptake was observed in the CFL market. This uptake resulted partly from the demonstration effect of phase 1 of the ELIB program.

ii. **June 2018:** The government requested additional financing II (AF II – IDA Credit 6202) to significantly scale up the project, given the achievements of most targets ahead of schedule. The change in results framework and the extension of the project's closing date were natural consequences of the mobilization of this additional financing.

iii. **May 2021:** The extension of the closing date of the project by two years (from December 2021 to December 2023) was decided to (i) enable the utilization of the newly approved Green Climate Fund grant (approved in 2020) to achieve the objectives of component B, and (ii) recover from the implementation delays resulting from COVID-19 pandemic restrictions.

II. OUTCOME

A. RELEVANCE OF PDOs

Assessment of Relevance of PDOs and Rating

4. **The relevance of the PDO at project closing is rated High**. This rating focuses on high-level objectives' relevance.

5. **The PDO is consistent with the current World Bank's Bangladesh Country Partnership Framework (CPF) for FY2023-2027[10], as well as 2011-2022 Bank strategies.** The current CPF includes the High-Level Outcome C (HLO C) of 'Enhanced climate and environmental resilience'. HLO C partly aims at 'scaling up renewable energy resource development' and 'improving efficiency in energy consumption in Bangladesh'. Within this framework, the CPF has listed RERED II as one of the ongoing operations helping the World Bank achieve these objectives in Bangladesh. It was aligned with (i) the Country Assistance Strategy (CAS) for FY2011-2015[11] under the strategic objective #1 of 'accelerated growth: increase transformative investment and enhance the business environment' through outcome 1.3. 'Increased infrastructure provision, access and efficiency', and (ii) the CPF for FY2016-2020[12], under Focus Area 1 'Growth and Competitiveness' with the Objective 1, 'Increased power generation capacity and access to clean energy'.

6. **The PDO remains aligned with the government's high-level development priorities and sectoral needs, expressed through its Eighth Five-Year Plan Strategy for FY2020-2025**, notably in the 'Strategic framework for the power sector' and its components on (i) 'enhancement of the exploitation of […] renewable resources […] Includ[ing] solar energy, biomass, and waste to power, where the core strategic goal will be to make the energy available at optimum rate to all consumers' as well as (ii) 'improvement of energy efficiency and conservation through demand side management'. In addition, the design of the RERED II capitalized on existing government initiatives such as: (i) the Solar Home System Program of Bangladesh, started in 2003 with the help of IDA and other development partners; (ii) the National Domestic Biogas and Manure Program (NDBMP), started in 2006; and (iii) the ELIB program, abovementioned. This demonstrates an existing local need to which the project has provided necessary means to scale up.

B. ACHIEVEMENT OF PDOs (EFFICACY)

Assessment of Achievement of Each Objective/Outcome

7. **The overall efficacy of the project is rated Substantial.** A split evaluation was carried out to evaluate the achievement of each objective/outcome under the project before and after the June 2014 Level 1 restructuring (leading to a narrowed PDO, *see section on 'Significant changes')*. This account draws on: World Bank and external M&E reports, third-party monitoring surveys, insights provided by IDCOL and Power Cell, Bank specialists and field visits.

Summary of PDO Efficacy Rating

Outcome	Pre-2014 Level 1 Restructuring	Post-2014 Level I Restructuring
1. Increased access to clean energy in rural areas through renewable energy	Modest	Substantial
2. Promotion of more efficient energy consumption	Negligible	-
Average Rating	**Modest**	**Substantial**

3. **These ratings evaluate achievement of each PDO-level outcome based on (i) the causal link observed between project investments and achievement of project objectives; and (ii) project achievement rates**.

Objective / Outcome 1: Increase access to clean energy in rural areas through renewable energy

Efficacy pre-June 2014 Restructuring: Modest. Efficacy post-June 2014 Restructuring: Substantial

29. **Before the restructuring, delays in the implementation of most sub-components during the first two years of the project adversely impacted the project's efficacy.** In 2012-2014, there was strong momentum for Component A.1. SHS scale-up, the most capital-intensive activity. However, all other activities lagged significantly until the June 2014 restructuring. While some delays were outside the team's control (such as the time-consuming processes of ministerial approvals for the Technical Assistance, or political turmoil in 2013-2014), other delays were directly attributable to the implementation of suboptimal modus operandi (SIP and solar mini-grid activities, *see 'Factors subject to Implementing agencies' control*').

Indicator - Outcome 1: "Increased access to clean energy in rural areas through renewable energy"		Unit	Achievement before June 2014 restructuring		
No.	Name	Unit	Appraisal target	Actual 6/28/2014	Achievement Percentage
PDO level					
1	Number of households, farmers, and businesses having access to clean energy services	Number (million)	1.58	0.42	26.37
2	Generation Capacity of Renewable Energy (other than hydropower) constructed	MW	61.00	13.95	22.87
3	Direct Project Beneficiaries	Number (million)	10.40	2.22	21.32
4	Female beneficiaries	Percentage	55.00	46.00	83.64
5	People provided with access to electricity under the project by household connections	Number (million)	2.50	2.02	80.68
Intermediate result indicator level					
6	Number of solar home systems installed	Number (million)	0.55	0.41	75.36
7	Number of connections made through mini-grid systems and captive plants	Number	6,750.00	-	-
8	Number of solar irrigation pumps	Number	1,500.00	8.00	0.53
9	Collection efficiency of the SHS POs	Percentage	N/A	N/A	N/A
10	Enabling policy for renewable energy development	Text	SREDA Operational with core staff hired	Not achieved	Not achieved
11	Number of female staff in Pos	Number	N/A	N/A	N/A
12	Grievances received that are addressed within two months of receipt	Percentage	N/A	N/A	N/A
N/A	**Achievement rating before June 2014 restructuring**		**Modest**		

Table 3: Outcome 1 achievement rates before June 2014 restructuring

[10] CPF for Bangladesh, FY2023-2027 (Report No: 181003-BD)
[11] CAS for Bangladesh, FY2011-2014, later extended to 2015 (Report No. 54615-BD)
[12] CPF for Bangladesh, FY2016-2020, later extended to 2022 (Report No. 103723-BD)

30. **Following the restructuring, IDA investments have achieved the intended objective of Outcome 1 by helping alleviate key barriers to the dissemination of renewable energy solutions in rural areas.** To achieve this objective, IDA funds (transiting through IDCOL) were used to nudge a variety of rural actors into transitioning to clean and renewable energy, a transition which would have not taken place without project intervention. IDA funds have contributed to removing the key barriers to disseminating these solutions, which were (i) the prohibitive cost of purchasing or operating these solutions for actors within this ecosystem (financiers, developers, operators, consumers); and (ii) needs to create an overarching institutional and policy environment. The project has enabled increased energy access (PDO-level outcome), thereby contributing to improving the livability of rural areas (long-term outcome) for households (with an emphasis on women) and businesses through several channels (increased livelihood, better health, decreased out-of-pocket energy expenditure, etc.).

1. **More specifically, IDA funds were used to lay a foundation for the widespread adoption of these solutions, by:**

(i) Stimulating a demand for these solutions, by
   a. Bringing down their cost of adoption for households through (a) direct grant subsidies to customers, and (b) financial incentives to POs and MFIs to finance projects they would not have supported otherwise.
   b. Financing awareness-raising campaigns and robust testing and certification of solutions distributed[14]. By helping improve product quality, the project increased trust and demand for more modern cooking energy solutions, despite their higher cost compared to readily available energy solutions like burning wood[15].

(ii) Stimulating a supply for these solutions, by
   a. Creating an enabling environment for the development of renewable energy in rural areas in Bangladesh, through technical assistance that supported the creation of key institutions (SREDA), plans (such as the Renewable Energy Development Plan), and extensive technical-level rules, acts, guidelines, and trainings[16]. Annex 6.1. provides a detailed list of TA activities and outcomes.
   b. Encouraging IDCOL in supporting POs in becoming commercially viable entities operating in the field of clean and renewable energy development ('entrepreneurship' capacity building).

2. **This market development support resulted in substantial achievement rates, while some challenges can be noted.** After the restructuring, the project nearly achieved all its PDO and intermediate result targets for Outcome 1, with underachievement on some indicators (see Table 4) due to two main factors. *First*, the key assumption of the project (slow grid expansion) did not fully materialize as the government engaged in parallel energy development policies (such as free distribution of SHS and accelerated grid expansion, supported by the World Bank policy dialogue with the GoB on energy) which encouraged the early cessation of some activities (SHS deployment and solar mini-grid development). This contributed to lower-than-expected achievement rates under these sub-components (such as the number of mini-grids installed), and PDO-level indicators (such as the number of beneficiaries, following a lower-than expected number

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[14] Such as the ICS Testing Facility at Department of Chemical Engineering of Bangladesh University of Engineering and Technology (BUET).

[15] The ICR mission collected evidence of such results beyond project life. Interviews with randomly sampled final customers indicated a strong willingness-to-pay for new clean cooking solutions, when current ones come to the end of their lifecycle. In addition, following the project closing, the number of ICS deployed rose to over 4 million.

[16] Power Cell reported 2580 man-hours local training provided each year from the project and about 4480 man-days foreign training conducted from the project for HR development of the utilities and project personnel.

of SHS beneficiaries effectively targeted). *Second*, the modus operandi envisioned for the dissemination of some energy solutions (mini-grid systems, biogas captive plants, and solar irrigation pumps) proved more difficult to implement than expected at appraisal[17]. While the team took corrective measures to address these challenges, they took time to yield results. This explains a slow uptake in the realization of intermediate results-level indicator 8 and 9 (Table 4), both in 2014 and in 2018. This delay was not entirely caught up by project closing in 2023, which ultimately led to a below-target number of solutions disseminated (intermediary result), and therefore to a below-target number of beneficiaries reached (PDO result).

Indicator - Outcome 1: Increased access to clean energy in rural areas through renewable energy		Unit	Achievement before June 2014 restructuring			Achievement after June 2014 restructuring, at the initial project closing (Dec 2018)			Achievement after June 2014 restructuring, at final project completion (Dec 2023)		
No.	Name	Unit	Appraisal target	Actual 6/28/2014	Achievement Percentage	2014 revised target	Actual 3/16/2018	Achievement Percentage	2018 revised target	Actual 12/18/2023	Achievement Percentage
PDO level											
1	Number of households, farmers, and businesses having access to clean energy services	Number (million)	1.58	0.42	26.37	2.07	2.38	114.80	6.09	5.28	86.77
2	Generation Capacity of Renewable Energy (other than hydropower) constructed	MW	61.00	13.95	22.87	56.00	49.00	87.50	88.00	85.00	96.59
3	Direct Project Beneficiaries	Number (million)	10.40	2.22	21.32	5.70	6.51	114.19	9.80	9.56	97.55
4	Female beneficiaries	Percentage	55.00	46.00	83.64	59.00	59.22	100.37	76.00	71.27	93.78
5	People provided with access to electricity under the project by household connections	Number (million)	2.50	2.02	80.68	4.70	5.42	115.32	5.50	5.43	98.80
6	People who gained access to more energy-efficient cooking and/or heating facilities	Number (million)	4.30	-	-	1.00	1.16	116.10	5.50	4.05	73.64
Intermediate result indicator level											
7	Number of solar home systems installed	Number (million)	0.55	0.41	75.36	1.03	1.20	116.72	1.20	1.20	100.23
8	Number of connections made through mini-grid systems and captive plants	Number	6,750.00	-	-	7,500.00	5,500.00	73.33	26,881.00	9,082.00	33.79
9	Number of solar irrigation pumps	Number	1,500.00	8.00	0.53	1,250.00	800.00	64.00	1,987.00	1,303.36	65.59
10	Collection efficiency of the SHS POs	Percentage	N/A	N/A	N/A	90.00	85.00	94.44	90.00	83.00	92.22
11	Number of improved cookstoves purchased by households	Number (million)	1.00	-	-	1.00	1.16	116.00	5.00	4.05	81.00
12	Number of biogas plants installed	Number	20,000.00	1,453.00	7.27	33,000.00	5,000.00	15.15	8,300.00	10,907.00	131.41
13	Enabling policy for renewable energy development	Text	SREDA Operational with core staff hired	Not achieved	Not achieved	SREDA Operational with core staff hired	Not achieved	Not achieved	SREDA Operational with core staff hired	SREDA Operational with core staff hired	100.00
14	Number of female staff in Pos	Number	N/A	N/A	N/A	N/A	N/A	N/A	480.00	2,321.00	483.54
15	Grievances received that are addressed within two months of receipt	Percentage	100.00	98.00	98.00						
N/A	Achievement rating across entire project time		Modest			Substantial			Substantial		
N/A	Achievement rating on average		Substantial								

Table 4: Outcome 1 achievement rates after June 2014 restructuring

[17] More information on this aspect is provided under the section "Key factors that affected Project Implementation".

Objective / Outcome 2: Promotion of more efficient energy consumption

Efficacy pre-June 2014 Restructuring: Negligible. Efficacy post-June 2014 Restructuring: N/A (deleted)

33. **The efficacy of the project with respect to the outcome 2 is judged negligible before the June 2014 restructuring, and not-rated after the June-2014 restructuring as this marked the outcome's deletion from the formal PDO.** The rationale to delete this outcome from the PDO can be challenged. Indeed, this outcome was deleted once the energy-efficient lighting activity (former Component C) was dropped. Yet, while energy-efficient lighting is explicitly labeled as "energy efficient" under the results framework, the project comprises two other solutions that can also be labeled explicitly energy-efficient: improved cookstoves and domestic biogas digesters, as these appliances' salient feature is to use less biomass than a traditional way of cooking, to achieve greater fuel and heat efficiency. Therefore, before the 2014 restructuring, the ICR will assess the efficacy of Component B (Household Energy) under Outcome 2.

34. **Before the restructuring, the project's performance on this outcome is negligible.** To achieve outcome 2, the investments were designed to promote more efficient lighting and cooking energy consumption, by using IDA funds to encourage the creation of a strong market for energy efficient appliances. On the cooking segment, little to no progress was made, as reflected by very low achievement rates (see Table 5). This was due to a long preparatory and testing phase for ICS, and inadequate choice of technology for DBD[18]. On the lighting segment, CFL was abandoned as market evidence suggested that the uptake was already in effect without concessional financing.

Indicator - Outcome 2: "Promotion of more efficient energy consumption"		Unit	Performance before the June 2014 restructuring		
No.	Name	Unit	Appraisal target	Actual 6/28/2014	Achievement Percentage
PDO level					
1	More efficient energy consumption through introduction of energy-efficient lighting	MW saved per year	160.00	-	-
2	Direct Project Beneficiaries	Number (million)	10.40	2.22	21.32
3	Female beneficiaries	Percentage	55.00	46.00	83.64
4	People who gained access to more energy-efficient cooking and/or heating facilities	Number (million)	4.30	-	-
Intermediate result indicator level					
5	Number of improved cookstoves purchased by households	Number (million)	1.00	-	-
6	Number of biogas plants installed	Number	20,000.00	1,453.00	7.27
7	Number of energy efficient lamps distributed	Number (million)	7.25	-	-
N/A	**Achievement rating on average**		**Negligible**		

Table 5: Outcome 2 achievement rates before June 2014 restructuring

[18]Corrective measures (change of technology and readjustment of target) allowed this activity to overperform by closing (see outcome 1). See section 'Key factors that affected Project Implementation' for more information on corrective measures.

***Other*. Commercially viable market development**

5. **Although it was not an explicit project objective, the design of the project aimed at encouraging the creation of a "commercially viable market" for the suggested solutions. The ICR assesses that a commercially viable market was partially encouraged by the project, but not fully established.** IDA funds were used to scale up the uptake of the proposed energy solutions by de-risking the investments taken by MFIs and private operators, progressively reducing the subsidization of the proposed solutions for larger scale POs (from an average level of 90% of subsidization per solution to 20% at closing, for the largest POs and on some solutions), encouraging new business models and market entry of new actors. While no evidence allows to conclude that the RERED II has enabled POs to reach full financial viability[19], evidence suggests that the project has laid the foundation to initiate the next stage for this market – that is, full commercial uptake (see figure 9) – especially for SIP and clean cooking solutions. On SHS, market evolution was characterized at closing by saturation declining adoption trends, with 2.25 percent of the population using SHS in 2021, and 1.81 percent in 2023[20]. On market outlook, former SREDA chairman declared "there is no future for SHSs as electricity has been made available in rural areas", with a national electrification rate of 97.54 percent at closing.

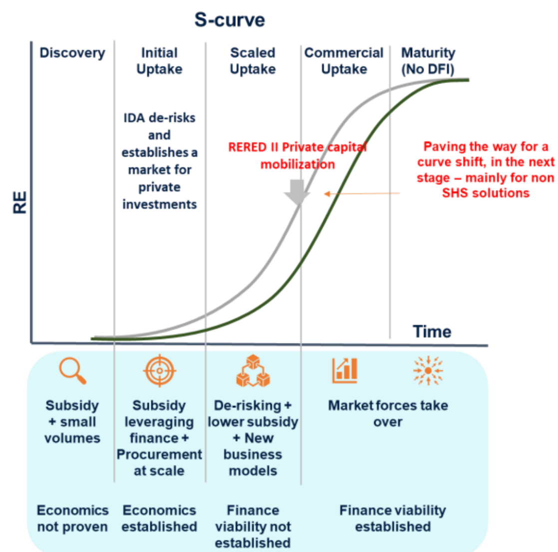

Figure 9. RERED II has encouraged the scaled uptake of off-grid clean energy solutions in rural areas

Justification of Overall Efficacy Rating

6. **The overall efficacy is rated Substantial.** Due to a timely restructuring in 2014, upon completion in 2023, all PDO targets and intermediary targets were nearly achieved, achieved or overachieved. At completion, the project had benefited 12.9 percent of the households in Bangladesh and provided energy access to 19 percent of its rural households[21]. The project successfully installed 85 MW[22] of renewable energy capacity (mainly solar), against a target of 88 MW. Per capita

[19] IDCOL and other MDBs such as the Asian Development Bank and KfW are still supporting the next generation of repeater projects through grants to ascertain the scaled uptake (for instance of SIP), prior to letting market forces fully take over (e.g. 0% subsidies).
[2020] Bangladesh Bureau Of Statistics, 2023.
[21] At project closing, the population of Bangladesh was 170 million, composed of 41 million households, out of which 27.8 million households were distributed in rural areas. (Statistical Yearbook, Bangladesh, 2022, page 22). At project closing, out of the 5,283,323 final beneficiaries having access to clean energy services (results framework indicator), 5,260,956 were households, 11,799 were businesses, and 10,568 were institutes.
[22] This represents 18.5% of the installed solar PV capacity (459 MW) of Bangladesh, Bangladesh Power Development Board, 2022-23.

electricity consumption almost doubled to 464 kWh per year from approval to closing, in part thanks to the project. These achievements were made possible by the use of IDA funds, which helped to lift some of the main barriers to the adoption of renewable energy solutions in rural areas.

C. EFFICIENCY

Assessment of Efficiency and Rating

7. **Economic efficiency. Rating: Substantial.** The majority of the project's activities are estimated to yield healthy economic and/or financial rates of return. The sole exception is the mini-grid program, which is significantly hampered by high capital costs. Additionally, the efficient fluorescent lighting program was entirely abandoned, signaling a desire to reallocate funds to a better 'value for money' use. For details, see the Summary table below, and Annex 4.

8. **Comparison of the ex-post and ex-ante economic analyses:** The economic analysis at appraisal found all subcomponents to have healthy rates of return. At closing, however, the results proved slightly different. Though the Household Energy component and the SIP component exceeded appraisal expectations, the same cannot be said about the remaining segments of RAPSS, Mini-Grid and Bio-Electricity. Both are hampered by significantly higher capital costs than initially expected at appraisal. The Summary table below and Annex 4 provide further details.

9. **Summary of results & Sensitivity analysis:** The table below summarizes the economic and financial returns for various subcomponents of RERED II at both appraisal and closing stages. See Annex 4 for additional details.

	Stand-alone		RAPSS						Household Energy			
	SHS		Solar Mini-Grid		Solar Irrigation Pump		Gasifier/Bio-Electricity Power		ICS		Biogas Plant	
	Appraisal	ICR	Appraisal	ICR	Appraisal	ICR	Appraisal	ICR	Appraisal	ICR	Appraisal	ICR
EIRR %	43	49							36.3	> 1000	27.3	29
FIRR %	26	46	13.7	1	10-13	11-13	28	12.3	91	441	15.1	50
Cost effectiveness												
Renewable Energy (BDT/m3			30.8	51.1	1.7	1	12	23.8				
Diesel Alternative (BDT/m3)			42.7	9.5	2.4	2.7	25	24.3				
Sensitivities	Kerosene use must drop to one liter per household for EIRR to drop to 10%.	Kerosene use must drop to 20 liters per year for all households for EIRR to drop to 10%.	PV system cost must be 33% higher to reach breakeven point.	PV system cost must be infinately lower to reach breakeven point.	Diesel Fuel Price must drop to 49.3 BDT/liter to break even.	To break even, Diesel Fuel Price must be 0.5% of its current value of across the years.	To breakeven, gasifier diesel fuel cost would have to reach 61 BTD/liter or 27.6 BTD/liter for the diesel generator. Otherwise biomass fuel would have to be 13.3 BDT/kg.	(Information breaking down biogas project's fuel usage unavailable, only general OPEX values provided)	If biomass fuel cost is 80% of its current cost, EIRR drops to 10%	If biomass fuel cost decreases by 93% of its reference cost in 2022, EIRR drops to 10%	If biomass fuel cost is 60% of its current cost, EIRR drops to 10%	If biomass fuel cost for households making over 10,000 BDT a year decreases by 90% of its reference cost in 2021, EIRR drops to 10%
Comments	The SHS program eventually delivered more than four times the amount of systems anticipated at appraisal. Additionally, systems of size from 10 Wp to 300 Wp were deployed, far more variety than anticipated.		PV system cost during implementation proved signifcantly higher than anticipated at appraisal. Additionally, at appraisail the alternative to the PV mini-grid was deemed a diesel mini-grid. However, on-the-ground surveys have concluded that solar mini-grids would most likely replace household kerosene cost for lighting. Thus ultimately, the lack of a high capital cost in the counterfactual has proven severely detrimental in producing positive returns.		At appraisal, a better performing diesel pumps was considered. Ultimately however, the diesel IDCOL provided as basis at ICR was one about half as efficient. Meanwhile, the solar pump considered at appraisal was significantly less efficient than the one deployed by the program, which pumped close to 5 times more water.		The greater distance between counterfactual and project levelized costs of electricity at appraisal and that at closing is ultimately explained by significantly higher project capital costs than anticipated.		As was even stated at appraisal, biomass fuel values were heavily undervalued in that analysis. More recent survey data suggest far more substantial household expenditure.		Excluded from the financial analysis are Project Investment costs (Grant and technical assistance which, amongst other things, promoted biogas plant use to the population), which were significantly more costly than anticipated at appraisal. So much so that at closure, these costs overwhelm the combined loss of economic health and domestic labor benefits, which are similarly not included in the financial analysis.	

Implementation efficiency – Rating: Substantial

40. **Implementation efficiency is deemed substantial at project closing, facilitated by a flexible and robust implementation arrangement and a strong coordination between stakeholders.** On paper, the project took almost twice as long to implement as envisaged at appraisal (11 years compared to an initial estimate of 6 years). However, this delay was not due to implementation efficiency challenges. On the contrary, an evaluation of project disbursements and project progress (Table 4) indicates that the project was extended twice to augment most of targets following a successful test-and-scale phase, accompanied by additional financing mobilization. Indeed, a few months before the first closing date extension from December 2018 to December 2021, the disbursement rates of IDA credit and grants were 90 percent and 92 percent, respectively[23]. Similarly, in September 2021, the IDA credit disbursement rate was 83 percent. By project closing, the disbursement rates of the IDA credits and project grants stood at 99.4 percent and 91.05 percent, respectively.

41. **Efficiency rating: The overall efficiency rating is Substantial**. At the economic efficiency level, the project's economic returns and benefit-cost ratio were substantial, with IRRs mostly improved compared to the estimates at the time of the project appraisal. At the implementation efficiency level, strong client ownership, supported by close Bank coordination, fostered efficient implementation.

D. JUSTIFICATION OF OVERALL OUTCOME RATING

42. **The overall outcome rating is Satisfactory.** The PDO was highly relevant at the time of project design and remains highly relevant to the Bank's strategy at project closure. At closing, most outcomes were almost fully achieved or exceeded, and the efficiency of the project is higher than estimated at design. This conclusion is also reflected in the awards received by the project from external third parties (see Annex 6.1.).

		Before June 2014 restructuring	**After June 2014 restructuring**
	Relevance of objective	High	
	Efficacy (PDO)	Modest	Substantial
	Outcome 1	*Modest*	*Substantial*
	Outcome 2	*Negligible*	-
	Efficiency	Substantial	
1	Outcome ratings	Moderately Unsatisfactory	Satisfactory
2	Numerical value of the outcome ratings	3	5
3	Disbursement	17.7%	82.3%
4	Weighted value of outcome rating	0.53	4.1
5	Final outcome rating	Satisfactory (4.63[24])	

[23] March 2018 ISR

[24] The rating is rounded to the numerical value of '5' to reach 'Satisfactory', considering the multifaceted achievements of the project.

E. OTHER OUTCOMES AND IMPACTS (IF ANY)

Gender

3. **The Project had a positive impact on gender equality through several channels**. *First*, women represented a large share of the final beneficiaries under the project, comprising nearly 70 percent of the final beneficiaries. By having access to clean renewable electricity and modern cooking energy, women have benefited from improved living conditions resulting in better health, increased safety (reduced fire hazard) and greater economic empowerment (income-generating opportunities such as knitting). *Second*, women represented an outstanding share of the project's workforce. The target of including at least 480 women as PO staff was achieved at a much broader scale than expected, with 2,321 women employed in POs (especially under ICS). Such achievements are significant in the context of rural areas in Bangladesh.

Institutional Strengthening

4. **The technical assistance component has had a positive trickle-down and multiplier effect on the institutional landscape for renewable energy in Bangladesh.** Indeed, the establishment of SREDA and the sector reform support carried out under the TA has enabled several other institutions to inform their ongoing renewable energy projects. This was enabled by a high number of studies supported by the TA, including consultancy services and other sector and technical works listed in Annex.6.1. of this report, along with a description of their corresponding outcomes.

Mobilizing private sector finance

5. **The Project relied on the mobilization of microfinance institutions (MFI), which have been leveraged thanks to several resources of external financing** (IDCOL, IDA, other development partners). In total, for each one US dollar of IDA financing used, POs have mobilized 0.44 US dollar (See '*Datasheet'*).

Poverty Reduction and Shared Prosperity

6. **First, some of the energy solutions promoted under the project have allowed beneficiaries to significantly reduce their out-of-pocket energy expenditure.** This is particularly evident in activities that allowed for the switch from fossil-fuel-based energy generation to cleaner energy generation, such as: (a) under the bioelectricity sub-projects where electricity produced from poultry litter/cow -dung has enabled a reduction in spending on diesel and electricity bills; (b) under SIP where the use of solar-powered irrigation resulted in time savings (50 percent less time-consuming than operating a diesel pump) and cost savings (irrigation costs are lower by 20-25 percent compared to using a diesel pump, with little to no need to pay for additional electricity). Farmers have also increased their earning opportunities, as SIPs have allowed the introduction of new crops, leading to an average increase in revenue by 19 percent.

7. **Second, this project has contributed to generating additional employment opportunities.** These jobs cover a wide range of skills. In terms of direct employment opportunities, 1,300 direct jobs were created under the SIP sub-project, through the creation of the position of 'private operator' (one per SIP), locally hired and paid by the PO. In indirect employment opportunities, almost all of the energy solutions distributed under component A and component B were either fully or partially locally manufactured. Under the ICS sub-project, 10,000 indirect jobs were created. Additionally, indirect self-employment opportunities were also facilitated thanks to energy and grid-quality electricity provided to individuals and businesses.

Other Unintended Outcomes and Impacts

48. **The project has had significant positive environmental effects, although these effects were not directly monitored under the results framework**. Indeed, due to the transition from heavily polluting primary energy sources to cleaner alternatives, the Project has directly contributed to a reduction in harmful GHG emissions and other air pollutants. For instance, under ICS, a study[25] found that ICS dissemination reduced beneficiaries' exposure to PM2.5 and carbon dioxide by 31 percent and 86 percent, respectively, compared with traditional stoves originally used. Due to its significant environmental benefits, the ICS was registered under the Clean Development Mechanism (CDM) in 2018 and was able to issue 1.78 million certified emission reductions (CERs). The TA has facilitated clean energy generation projects through feasibility studies on solar and wind power plants at Sonagazi, Feni; Rangunia, Chattogram). In addition, CFL and battery disposal/recycling guidelines have been prepared and implemented. The project also helped indirectly establish 3 battery recycling facilities (for batteries used under solar projects, but also for vehicles).

49. **The project also generated key technical improvements that benefited the Bangladeshi renewable energy ecosystem, an achievement reflected in the awards received by the project (see Annex 6.i.)**. *First*, the quality, performance, and efficiency of several solutions was enhanced through the establishment of technical committees that designed specific standards to be met by the disseminated solutions. *Second,* the experience of SIP and ICS deployment directly informed the Government of Bangladesh's policy on irrigation[26] and clean cooking.[27] *Third,* the project generated carbon credits under component B (ICS), in compliance with Article 6 of the Paris Agreement, generating additional revenues for the project (ultimately shared between IDCOL and the POs) and helping Bangladesh pursue voluntary cooperation in implementing its nationally determined contribution (NDC).

III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

Assessment of the project design and background analysis

50. **When designing the Project and setting its targets, the level of ambition for RERED II could have been further aligned with the trends observed under the RERED I project**. Notably, the majority of targets were either nearly met or exceeded on two occasions, close to the completion date of the project (see Table 4). This observation encouraged the teams to further increase the level of ambition for the RERED II, as reflected by the changes in the results framework and mobilization of two successive additional credits. This suggests that there may have been potential to further increase the original ambition during preparation, had it further relied on the predecessor's project's findings on key elements such as willingness to pay for assets and services, sustainability of proposed schemes and behavioral changes receptive to the energy transition. In addition, the premature closing of Component C 'Energy Efficient Lighting' only two years after the Project's approval due to market irrelevance could have been avoided if thorough market studies and field-level surveys on CFL adoption by the targeted populations had been conducted during preparation. Despite this, strong Government ownership and the desire to maintain this activity resulted in its inclusion in the project design, although it was subsequently dropped early.

[25] Berkeley Air Monitoring Group study conducted in 2023, see Annex 6.1.
[26] Guidelines for the Grid Integration of Solar Irrigation Pumps (2020), Power Cell
[27] Bangladesh Country Action Plan for Clean Cookstoves (CAP) Hyperlink: *Bangladesh Country Action Plan (CAP) | Clean Cooking Alliance*

B. KEY FACTORS DURING IMPLEMENTATION

Factors outside of the Project's control

51. **COVID-19 pandemic, movement restriction and commodity price volatility:** The COVID-19 lockdown and associated movement restrictions in project areas adversely impacted the ability of POs to install equipment on time, as well as the roll-out of feasibility studies carried out under the TA. Consequently, this affected the timely completion of field-level activities – including outreach and capacity building. The pandemic and its supply chain disruption also had an adverse effect on the volatility of prices of raw materials essential for the production and maintenance of the developed energy solutions (such as metals and minerals); this effect was further exacerbated by Russia's invasion of Ukraine. The impact of Covid-19 may also have adversely affected business operations, leading to unforeseen delays in the mobilization and effective disbursement of grant funding for the ICS deployment.

52. **Political unrest during the early stage of project implementation (2013-2014):** In 2013-2014, Bangladesh experienced episodes of political unrest of different nature, marked by successive strikes, episodes of violence, road blockades and overall disruptions of the socio-economic circuit. These events have significantly slowed the pace of implementation until 2014, especially for activities involving local production and supply chains. POs faced significant challenges in supplying materials to centers involved in product development and testing due to these challenges.

Factors subject to the Bank's control

53. **Adjustment of project's ambition level:** IDCOL and the World Bank proactively monitored systemic trends during implementation, flagging them clearly during Implementation support missions (ISMs), as reflected in overall project and component-wise restructurings. As a result, restructurings and mobilization of additional credits were responsive to both beneficiary needs and the client's ability to deliver. Finally, bi-yearly technical missions with subject matter experts, coupled with randomized monitoring of project areas, provided alerts to any contractual, implementation or technical deviations and suggested corrective actions when relevant.

Factors subject to the Government's control

54. **Accelerated grid expansion in project areas:** Starting from 2015, the GoB accelerated the expansion of the electrical grid in project areas through rural cooperatives under the Bangladesh Rural Electrification Board (BREB). This accelerated expansion aimed at covering 90 percent of rural areas by 2018, with more than 300,000 connections realized per month. The project had not anticipated such a sustained and accelerated pace of grid expansion, potentially rendering some of its interventions obsolete earlier than expected. However, RERED II proved flexible to such circumstances, by (i) adapting its offerings: such as training POs to provide additional services like repair and replacement; and (ii) by mitigating the effects of grid expansion on POs: such as negotiating relaxed loan conditions, compensation mechanisms for affected POs and cooptation of POs to act as agents of Government policies on solar solutions (notably by relying on POs to help a Government policy on free distribution of SHS, which was carried in parallel to the RERED II). In addition, customer surveys demonstrated that despite grid expansion, targeted populations preferred to retain some energy solutions disseminated under RERED II as backups.

Factors subject to the implementing agencies' control

55. **Original selection of suboptimal modus operandi or equipment:**

(a) ***Solar Irrigation Pump (Component A.2. RAPSS):*** For the fee-for-service model to function effectively, SIPs needed to be installed on land parcels of 25 to 30 acres and required the cooperation of 20-25 farmers to be technically and economically efficient. However, such conditions proved scarcer than predicted at appraisal. While a more thorough preparatory phase could have allowed for a better evaluation of this aspect, IDCOL explored the feasibility of SIP ownership by individual farmers with smaller landholdings during implementation. This option was not retained due to affordability challenges. Yet, the SIP deployment gained momentum in 2015 as more suitable sites were identified.

(b) ***Solar Mini-grids (Component A.2. RAPSS)***: Identifying suitable lands for the installation of solar mini-grids proved more challenging than anticipated. In addition, a lengthy appraisal process for mini-grid projects placed significant emphasis on the POs ’role in identifying, testing and submitting funding proposals to IDCOL. Due to the POs ’capacity, this modus operandi proved suboptimal and triggered a delay in the deployment of this activity. In 2015, IDCOL explored an alternative approach where IDCOL would oversee site-specific searches, feasibility studies, and bidding. This proved efficient, and the activity gained momentum by mid-2015.

(c) ***Domestic biogas digesters for cooking (Component B. Household Energy):*** Traditional fixed-dome brick biogas plants were found to be inefficient as their installation took time, was not possible during rainy reasons, and varied in quality depending on masons. While this issue could have been identified during a preparatory and testing phase, IDCOL took the initiative to replace the model during implementation, prioritizing quality over quantity of cookstoves disseminated. Prefabricated biogas digesters, although more expensive, were preferred. While the switch to another technology took longer than anticipated, this corrective measure proved efficient as this activity picked up in 2018.

IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

M&E Design

6. **The results framework (RF) was partially adequate for monitoring the progress of project activity as several elements could have been adjusted to fully capture the RERED II achievements**. These include:

(i) <u>**Output-orientation**</u>: The results framework was largely output-oriented rather than outcome-oriented. While several target adjustments were made during restructuring, the indicators themselves were not revised to reflect best practices. This applies to all intermediate results indicators, which focus on number of solutions deployed instead of their quality or satisfactory level of functioning. In addition, the TA component’s indicator ‘Enabling policy for renewable energy development’ was measured against the operationalization of SREDA and the appointment of its core staff. While this is an important step towards the envisioned outcome, it is not sufficient to measure its full achievement.

(ii) <u>**Measurement of other key outcomes of the project**</u>: While environmental benefits and GHG emissions reduction are a critical aspect of the project’s outcomes, these were not reflected in the RF. Similarly, no indicator was introduced to capture the development of a commercially viable market or a private sector-led ecosystem for renewable energy, to capture the uptake in solutions proposed beyond the project closing.

M&E Implementation

7. **The impact and performance of the project were *monitored* through a combination of monitoring techniques employed by the Bank and the implementing agencies, adapted to the POs capacity and rural context of the project areas.** Outcome and intermediate results indicators were reported via quarterly financial monitoring reports (FMRs) submitted under the project. To inform these reports, IDCOL relied on a broad network of decentralized management and inspection units to conduct monthly randomized inspections of representative samples of portfolio projects under each sub-component. The results were encoded in an MIS software for all sub-projects, except under SIP and SMG, where manual reporting was necessary due to specific sub-project requirements. A thorough mechanism for customer-centered reporting was established through call-centers and quality customer satisfaction services[28]. Power Cell relied on a Microsoft Excel-based monitoring system to track the progress of its TA activities. No data errors or significant quality monitoring issues were reported during project implementation, a conclusion corroborated by the Mid-Term Review of the project and the ICR survey.

8. **The project impact was *evaluated* through a comprehensive range of impact evaluation techniques.** Several impact evaluation studies were carried out (See Annex.6.i). Studies were complemented by surveys (both field-based and phone-based), to measure the effect of the project. Surveys built on data gathered under the Multi-Tier Analysis for Electricity Access for Bangladesh[29], developed by the Energy Sector Management Assistance Program in 2015 – 2016.

9. **While the M&E implementation partially delivered its intended results, the ICR notes some possible improvements which would be implemented in future comparable projects in Bangladesh.** For instance, under the SIP activity, IDCOL is working with KfW to implement a digitalized monitoring and evaluation process that would cover future SIP deployment activities. Under Power Cell deployed activities, the government of Bangladesh is currently deploying a single window of financial management monitoring for all donor-funded projects (*see Financial Management)*.

M&E Utilization

0. **The M&E framework provided visibility on progress but only achieved a suboptimal performance as it did not allow the team to have a broad vision of outcome achievement.** Indeed, the M&E framework allowed for better guidance in implementation, scaling up or narrowing down decisions. For example, based on monitoring reports, under component A.ii. and B., a performance-based mechanism was introduced to encourage the deployment of energy solutions that were effectively used and well-functioning. Yet, it remained focused on access and did not provide sufficient tools to link the PDO, project activities and the modus operandi (PPP-based market development).

Justification of Overall Rating of Quality of M&E

1. **The overall rating of M&E is Modest.** This is mainly due to shortcomings in the M&E design which were not corrected during implementation and did not allow the M&E framework to reach its full utilization potential.

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[28] During the ICR mission, surveyed final participants reported high satisfaction of the quality of customer services, both in terms of means of communications (IDCOL call-centers, PO's direct contact, dedicated repair and maintenance staff) and in terms of quality of after-sale service (interventions in 24 to 48 hours, at no cost).

[29] The Global Multi-Tier Access Framework methodology measures access in terms of quality and affordability of electricity supply rather than measuring access in terms of binary definition (having or not having an electricity connection). This was developed by the Energy Sector Management Assistance Program (ESMAP) and is being rolled out in countries around the world in support of the UN Sustainable Energy For All (SE4All) initiative.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

Environmental and Social (E&S)

62. **E&S Compliance is assessed as Satisfactory.** The project was classified as Environmental Category B – Partial Assessment, which was an appropriate classification. Indeed, the main environmental concerns were relatively limited (mainly pertaining to contamination from improper SHS battery and CFL disposal), and adequate mitigation measures could be designed, in compliance with international standards. In addition, the project bore significant environmental benefits as activities contributed to reducing the consumption of polluting primary energy sources. At appraisal, it was assessed that the project triggered only one environmental safeguard policy: Environmental Assessment (OP/BP 4.01). Hence, the RERED II complied with the updated version of the Environmental and Social Management Framework (ESMF) developed for the RERED I, with provisions specifically adapted to the RERED II. Importantly, environmental management provisions were included in bid documents for sub-projects supported by RERED II. Recruitment and capacity building of E&S specialists were carried out satisfactorily[30]. No activities related to land acquisition, resettlement, relocation, or negative impacts on livelihood have been identified. Consultations with stakeholders have been carried. The Grievance Redress Mechanism (GRM) is active[31], and accessible.

Procurement

63. **Procurement is assessed as Satisfactory.** While procurement was rated at Substantial risk at appraisal, the RERED II designed and implemented adequate risk mitigation measures to address identified shortcomings. Indeed, the main risks stemmed from deficiencies including a shortage of staff trained on procurement matters in Power Cell, or lack of confidentiality during the bidding processes carried out by REB for Component C. 'Energy efficient Lighting' (which was dropped eventually). A salient feature of the project is that due to its nature of financial intermediary under the RERED II, IDCOL has not been involved in major procurement although it received the largest share of the World Bank financing. Therefore, the risk of commercial malpractices was at the sub-level of POs. Adequate mitigation measures included regular procurement audit by third parties to review PO procurement practices, capacity building (appointment of a Procurement Focal Point in each implementing agency, recruitment of a consultant to support Power Cell and IDCOL), planification and formalization of procurement strategies in collaboration with the Bank. Throughout project implementation, procurement performance was consistently rated Satisfactory by the World Bank. The only exception took place during a downgrade to Moderately Satisfactory in 2021, primarily on the account of shortcomings by both IDCOL and Power Cell in maintaining regularly update procurement plans and procurement information on the World Bank Systematic Tracking Exchanges in Procurement (STEP) portal. Owing to the continued efforts of both implementing agencies, the RERED II procurement rating remained Satisfactory until project closing.

Financial Management

64. **Financial Management is assessed as Satisfactory.** While the Financial Management (FM) risks were deemed Moderate thanks to the experience of implementing agencies with Bank-financed projects, FM arrangements were adequate to satisfy the fiduciary requirements of IDA. The FM mechanism for the project was aligned with the government Project Accounting Manual. Accordingly, the 'Foreign Aided Project Audit Directorate' (FAPAD) regularly audited the project. In the case of IDCOL, this audit was augmented by an internal audit, and a corporate-level audit by a private auditor. IDCOL benefitted from a robust automated accounting system, which reduced the

[30] Combination of (i) in-house strengthened capacity regarding E&S matters (establishment of an Environment and Social Safeguards Management Unit – ESMMU – to institutionalize the E&S management in its operations; recruitment of a full-time environment staff member working with POs and battery manufacturers/suppliers to raise awareness on E&S issues); and (ii) recruitment of an additional environmental consultant to strengthen the ESMMU in its process of environmental assessment of sub-projects.

[31] Over 30,000 grievances have been received and addressed.

room for manipulations or errors. Power Cell carried out an Excel-based financial reporting, which is deemed as satisfactory, with no issue to be reported. FM was rated in the Satisfactory range throughout almost the project's duration, reflecting sound financial management practices by both implementing agencies. This was reflected in timely submission of financial statements and reports. Audit opinions were consistently unqualified, meaning that submitted statements were deemed to provide a true and fair view of the project's financial statement. The ICR notes that IDCOL's FM performance was challenged by the requirements of the RERED II closing process, which can be avoided in the future with further preparation on this phase and, if required, capacity strengthening on the requirements of a World Bank project closing. The project reached a level of disbursement of 97.98 percent at closing (US$ 288.42 M), the remaining balance (12.33 percent of allocation) was cancelled and refunded to the Bank.

C. BANK PERFORMANCE

Quality at entry

65. **Quality at entry includes a strong coordination with client needs and capacity and sound approach, but shortcomings can be identified in project preparation and design, notably in the form of insufficient preparatory studies and flaws in the results framework design.** A project preparation mission and several technical missions were conducted. Measures to assure quality at entry and associated risks were correctly identified, however, they could have been strengthened as discussed in this ICR. Quality at entry included a sound project concept and approach, adequate technical, financial, and economic analysis, strong client ownership, adequate consideration of environmental and social safeguards, fiduciary assessments, and high relevance to policy and institutional context. The RERED II also included an interesting "test-and-scale" approach, which allowed for IDCOL and the task team to focus implementation and financial efforts on the solutions that seemed to have a market uptake and drop those that were deemed irrelevant for RERED II intervention.

66. **While the ICR recognizes that the strength of the project's design features a "test-and-scale" approach, such an approach does not imply that all market tests should be carried during implementation, at the risk of jeopardizing the objective of commercially viable market creation.** Indeed, there exists an inconsistency between an objective to create a market for goods and services, and the absence of basic market studies at preparation stage, to be able to monitor the successful creation of such a market. Some basic preparatory market studies that could have been carried include (i) customer willingness-to-pay (for instance for CFL, which was dropped two years after implementation due to DFI irrelevance, or for more expensive but higher quality DBDs), (ii) analyses of geographic distribution and commercial profile of targeted populations (for instance for SIP to ascertain that the targeted areas would be financially and geographically relevant, as this may have proved challenging during implementation), (iii) basic technology tests of some suggested solutions (such as fixed dome digesters that proved inefficient to install after realizing the difficulties related to unreliable masonry services). Finally, in general, sizing the prospective market at preparation would have allowed to measure its expansion at closing, and hence gauge the effect of the project. This inconsistency can be attributed to an unclear definition of the objective of market creation both (a) during appraisal (as this objective is sparsely described in the PAD, clearly targeted in the design of the project, but not included in the theory of change), but also (b) during implementation, as the objective of market creation was not consistently recognized as an actual expected project outcome by successive leaderships. Had the RERED II clearly included such an objective in the results framework and under the monitoring and evaluation framework (for instance through indicators such as "decrease of the level of subsidization of the suggested solutions from X% to Y%"), the project would have benefitted from better visibility on the market dynamics that were at play during implementation, and hence, suggested market supportive corrective measures as needed.

Quality of Supervision

67. **Quality of supervision pertains to significant proactivity and reactivity of task teams.** Implementation support missions (ISMs) were conducted regularly, timely, and with a solution-oriented approach. The Bank conducted bi-yearly ISMs, several technical missions, and in-between-missions progress meetings which helped in early identification and resolution of emerging issues. The Bank appropriately identified project performance dynamics during restructurings and designed corresponding changes, consistently with the "test-and-scale" approach. Regarding broader policy trends that were identified (such as grid expansion and efficient lighting program discontinuity), the task teams tried either to discourage any action that seemed inconsistent with the RERED II (such as the free distribution of SHS in parallel to the Project), or to remain consistent with the Bank-level policy dialogue (as the Bank was supportive of grid expansion) while taking appropriate actions on the RERED II as a reactive measure (discontinuing the SHS component as of 2017-2018, dropping the CFL component).

68. **From a project reporting and documentation perspective, some minor disparities can be observed:**

- (i) Inconsistency in the level and quality of information reported on, with detailed reporting focused on capital-intensive activities, leaving others, such as mini-grids and bioelectricity, less documented. This also includes a discrepancy between restructurings effectively implemented and reported.
- (ii) The quality of reporting on the Grievance Redress Mechanism (GRM) was suboptimal with detailed reporting starting only in late 2021, two years before closing. However, given the customer-centered approach of this project, a functional GRM is key to ensure sustained customer trust and satisfaction.

Justification of Overall Rating of Bank Performance

69. **The overall rating of World Bank performance is Moderately satisfactory primarily due to shortcomings in project and RF preparation, and gaps in project reporting and documentation**. These were mitigated by Bank performance during implementation, evidenced by regular solution-oriented missions and strong engagement.

D. RISK TO DEVELOPMENT OUTCOME

70. **Ensuring financial and environmental sustainability of outcomes is subject to effective mitigation of two risks:** (i) lack of institutionalization of the renewable energy market upon the end of concessional financial support to POs; (ii) expansion of gas-based grid-electricity at a lower tariff than the renewable energy solutions proposed. These risks will require continuous policy efforts to ensure that outcomes are sustained beyond the project's closing. The first risk is already being mitigated by integrating alternative revenue streams to sub-projects, such as carbon credits or grid integration of SIPs[32], which may reduce the dependency on concessional financing and increase these projects' profitability. The second risk would be mitigated by escalating electricity tariffs in Bangladesh, implementing renewable-energy scale-up policies, and achieving economies of scale in the development of off-grid solutions.

[32] Under the SIP sub-project of the RERED II, private developers can sell back excess electricity to the grid under specific conditions. Indeed, 9 SIPs have been selected to serve as pilots for excess electricity grid integration, at a tariff that does not allow cost recovery yet. However, to cover for the gap between the received tariff and the cost of generation, dedicated grant funding has been made available by the International Water Management Institute for this project, in the framework of a Research and Development operation. These 9 pilots will inform a greater scale operation of SIP grid integration, whereby a specific trade arrangement would allow SIPs to sell their excess generation only above a certain cost-recovery threshold (potentially above 8.5 BDT/kWh).

V. LESSONS AND RECOMMENDATIONS

<u>Lesson 1</u> **– Projects structured as a "test-and-scale" approach, enabling complementary but independent activities, help foster client ownership, enhance impact, and guide efficient use of financial resources.** The "test-and-scale" approach allowed the RERED II to focus financial and implementation efforts on the activities that empirically demonstrated their market potential during implementation, while dropping activities that showed signs of irrelevance. This approach facilitates the decision-making process on mobilizing additional financing, as it provides specific evidence to substantiate the financing needs of successful activities requiring greater capital investments to scale up (SHS, SIP, ICS) while divesting from others. This approach also allowed to take some calculated risks, by integrating the potential need to change some of the elements decided at appraisal (such as DBD technology choice, CFL component dropping). These changes were facilitated by a clear compartmentation of the project's implementation structure. Indeed, the implementation arrangement of the RERED II project empowered implementing agencies to manage each activity through improved local systems. Establishing separate sub-PIUs and project leads dedicated to each sub-component increased project implementation efficacy and efficiency, as the implementation modality was designed in a clear segmentation without adding unnecessary layers of complexity.

However, while the strength of this design is its flexibility and capacity to adjust the ambitions of the project as it progresses, it can incentivize indefinite project extensions and/or restructurings if the project design is not sufficiently backed by preparation studies. In the case of the RERED II, successive major restructurings allowed the project to drop irrelevant sub-activities and mobilize additional financing as the test-and-scale approach bore results. However, this approach comes with a high transaction cost which can be mitigated to reduce the number of avoidable changes with further preparation and scoping studies (for instance, willingness-to-pay, market sizing, geospatial assessments, etc.). In addition, had the prospective market been scoped at appraisal, a clear target of market size could have been set, to provide the Borrower and the Bank with an indication of the optimal life duration and/or size of the project. Therefore, such implementation arrangements should be paired with a robustly informed market analysis/scoping reflected in project design, and a clear vision of the optimal project life duration to minimize the risk of over-aging projects becoming irrelevant.

<u>Lesson 2</u> **– A market-based approach should always be designed with appropriate market regulation measures.** The RERED II supported the scale-up phase of a nascent market but has not led it to commercial viability as intended (see section '*Efficacy*'). This can be attributed to an unclear vision regarding the market development objective, as it was not sufficiently and explicitly enacted as forming part of the theory of change (see section '*Bank Performance*'). Therefore, the project design did not sufficiently protect the nascent market supported under the RERED II from traditional market risks, such as:

(i) <u>Market distortion</u>: In the case of the RERED II, the development of a commercially viable market for SHS was hampered by the emergence of market distortive trends such as (a) the emergence of an informal market, providing similar services and goods at lower cost/quality, (b) the distribution of free SHS by the government during project implementation, in parallel to the RERED II. As the Bank became aware of these potential trends thanks to broader sectoral policy dialogue with the Government, the project was adjusted to accommodate this market distortion, as a reactive measure.

(ii) <u>Obsolescence</u>: In the case of the RERED II, the market deployment of some proposed solutions (SHS, mini grid) may have suffered from earlier-than-expected accelerated grid expansion. Unsustainable price competition or technological advances of alternative solutions can also fuel obsolescence.

(iii) <u>Financial risk</u>: a well-functioning goods and services market should be supported by a healthy financial market. In the case of the RERED II, this financial market (through the financial intermediation operated by the IDCOL/PO PPP) was not backed by robust buffers in case of default of final customers. Indeed, until 2015, the most capital-intensive activity (SHS) focused on portfolio expansion rather than portfolio quality, which may have adversely affected the ability of POs to maintain long-term financial viability. Due to market distortion and earlier-than-expected partial obsolescence, customers' willingness to reimburse microloans contracted for SHS purchase decreased. However, a design shortcoming aggravated this situation: the absence of a strong collateralization or security mechanism designed to back default cases. This has led to an increasingly alarming default rate, reflected in an underperformance of the SHS collection efficiency of POs (from 90 percent in 2013 to 7 percent in 2018). Some measures were designed to mitigate the effect, such as local collection campaigns, and eventually partial waivers of loan obligations on some customers[33].

For future operations, such risks should be mitigated both at project-level and at policy level:

(i) <u>*At the project level,*</u> the largest room for maneuver resides in the flexibility of the project design. If the project design allows to easily adapt project activities to changing circumstances, such risks can be partially mitigated, as was the case for the RERED II.

(ii) <u>*At policy level,*</u> the largest room for maneuver resides in the ability to design a consistent policy framework:

a. <u>Economic regulation:</u> To minimize the risk of emergence of an informal market selling poor quality appliances, economic regulation at Government level can suggest rules to define the market scope. For instance, market entry controls such as licenses can exclude actors that do not supply quality and safe appliances. A strong enforcement campaign would provide the necessary credibility and consistency to such measures.

b. <u>Social regulation:</u> To minimize the risk of emergence of an informal market selling poor quality appliances, technical and environmental standards can be set at the government level, requiring market actors to abide by certain standards (safety, emissions, recycling protocol). Enforcement efforts would also be key.

c. <u>Financial regulation and rules</u>**:** To support a healthy financial market that would sustainably accompany the growth of a goods and services market, it is essential to design prudential regulations at the government level and financial rules at implementing entities' level to ensure significant solvency and liquidity buffers of financiers (tailored ratios, collateralization mechanisms). Additional financial safety nets in case of default can also include credit enhancement products such as insurances and guarantees (as was the case for some of the solutions proposed under the RERED II), if the scale of the project allows reducing the cost of these products. A more holistic approach to customer default can include an ex-ante designed protocol for collection post-default, as well as enhanced payment facilities. Finally, debt relief plans and loan waivers can be part of the financial regulation framework as an exceptional and partial solution, but should not be the rule.

Such regulations can be eased or lifted upon the establishment of a more robust market.

<u>Lesson 3</u> – Given the relatively fast-paced change in socioeconomic, infrastructural, and technological contexts, off-grid electrification efforts should allow the possibility of providing energy access through a service, a good

[33] In June 2022, out of a total loan amount of BDT 4,545.03 Crore (disbursed to 53 POs), the GoB and IDCOL Board approved a waiver of BDT 278.74 Crore of POs loans. Another BDT 105.38 Crore PO loans were written-off due to various non-compliance issues.

and/or design it as a revenue generating mechanism. Indeed, some financial models based on asset ownership may prove relevant during the early stages of implementation, but no longer during implementation as the context changes. For the RERED II, at appraisal, the SHS purchase model seemed the most relevant in a context where SHS was the only energy access option in the near to medium-term. However, as implementation progressed, and as other energy access options were introduced faster than anticipated (such as the grid), owning the SHS made less sense for users who only wished to keep it as a back-up. Allowing an alternative model such as the 'pay-as-you-go' model (whereby SHS are owned by operators and final customers only pay a fee per use) may be more adapted and help control for financial risks related to debt-for-assets models. Similarly, off-grid electrification projects with high initial capital investments required from developers should be accompanied by a diversified set of financial risk mitigation strategies, to account for the risk of costly stranded assets as the infrastructural and technological landscape changes. For instance, under the RERED II, the discrepancy between anticipated and actual capital costs proved significant for the RAPSS mini-grid component, resulting at closing in a reversal of the positive economic viability observed at appraisal. To encourage the development of such risky projects for sponsors, some risk mitigation measures can include: (i) appraisal processes for mini-grid sites that would be standardized at implementing-agency-level (and not at developer-level), (ii) design of ex-ante financing strategies to support grid integration at the lowest cost if the grid was to reach the project area, (iii) maximize the possibility of revenue generation to increase the profitability of projects (be it through grid integration when possible; through Carbon Credits generation as was done under the ICS component, etc).

In addition, to better predict the changes that a project would face, it is crucial to collect more accurate on-ground data at appraisal. It is understandable that at appraisal, a time characterized by uncertainty, anticipated costs and assumptions are always unlikely to fully match the reality. In future endeavors, it would be more efficient to collect as much as possible on-ground data through partners and surveys, as well as consider a wider scope of sensitivities, to better prepare for the realities observed during implementation.

Lesson 4 – To optimize their impact, technical assistance activities should be designed to encourage transferable knowledge and skillsets. In the case of the RERED II, non-lending activities carried by Power Cell, such as the analytical work and policy design enabled by the project's technical assistance under Power Cell, have been an extremely efficient and high-yield tool under this engagement, as a relatively limited envelope (US$ 5 million at appraisal, US$ 13.1 million at closing[34]) led to significant outcomes[35]. Under the RERED II, the technical assistance led to positive trickle-down effects, which helped coordinate and scale up overall support to the government programs and build consensus on challenges and policy options, through thorough research and analysis. In addition, the technical assistance activities carried under the IDCOL-implemented components also helped foster local markets (ICS, batteries, DBDs, etc.) and encourage new and transferable skillsets, transforming POs into service providers and operators for renewable energy services (notably in solar power). This would help the expansion of renewable energy beyond the specific segments of SHS and SIP and would be used to expand utility-scale solar power and implement the energy transition of Bangladesh. However, impacts from RERED II non-lending activities were not explicitly measured by the results framework of the project. The next similar TA or projects should include design features to make the impact of critical analytical work more explicit, hence appropriately valuing its benefits.

[34] At closing, the fund allocation to Power Cell had reached SDR 9,939,441 (or USD 13.1 million, at the SDR:USD valuation dating May 15, 2024). The envelope increase follows the reallocation of funding between disbursement categories described in the section "Main changes".
[35] See detailed list of outcomes in Annex 6.2.

ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS INDICATORS

A.1 PDO Indicators

Objective/Outcome: To increase access to clean energy in rural areas through renewable energy

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of households, farmers, and businesses having access to clean energy services	Number	0.00 15-Jun-2012	1,578,000.00 31-Dec-2018	6,085,082.00	5,283,323.00 18-Dec-2023

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Generation Capacity of Renewable Energy (other than hydropower) constructed	Megawatt	0.00 15-Jun-2012	61.00 31-Dec-2018	88.00	85.00 18-Dec-2023

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Direct project beneficiaries	Number	0.00 15-Jun-2012	10,400,000.00 31-Dec-2018	9,800,000.00	9,563,900.00 18-Dec-2023

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Female beneficiaries	Percentage	0.00 15-Jun-2012	55.00 31-Dec-2018	76.00	71.27 18-Dec-2023

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
People provided with access to electricity under the project by household connections	Number	0.00 15-Jun-2012	2,500,000.00 31-Dec-2018	5,500,000.00	5,433,926.00 18-Dec-2023

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
People who gained access to more energy-efficient cooking and/or heating facilities	Number	0.00 15-Jun-2012	4,300,000.00 31-Dec-2018	5,500,000.00	4,054,758.00 18-Dec-2023

Comments (achievements against targets):

A.2 Intermediate Results Indicators

Component: Access to Electricity

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of solar home systems installed	Number	0.00 31-Dec-2012	550,000.00 31-Dec-2018	1,200,000.00	1,204,912.00 30-Jun-2021

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion

Number of connections made through mini-grid systems and captive plants	Number	0.00 31-Dec-2012	6,750.00 31-Dec-2018	26,881.00	9,082.00 18-Dec-2023

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Collection efficiency of the SHS POs	Percentage	0.00 28-Feb-2014	90.00 31-Dec-2018		83.00 31-Dec-2021

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of solar irrigation pumps	Number	0.00 31-Dec-2012	1,500.00 31-Dec-2018	1,987.00	1,303.00 18-Dec-2023

Comments (achievements against targets):

Component: Household Energy

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of biogas plants for cooking	Number	0.00 31-Dec-2012	20,000.00 31-Dec-2018	8,300.00	10,907.00 18-Dec-2023

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of higher efficiency cook stoves disseminated	Number	0.00 31-Dec-2012	1,000,000.00 31-Dec-2018	5,000,000.00	4,054,758.00 18-Dec-2023

Comments (achievements against targets):

Component: Sector Technical Assistance

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Enabling policy for renewable energy development	Text	SREDA not operational 30-Jun-2012	SREDA operational with core staff appointed 31-Dec-2018		SREDA operational with core staff appointed 30-Sep-2020

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Number of Female Staff in POs	Number	282.00 30-Nov-2017	480.00 18-Dec-2023		2,321.00 18-Dec-2023

Comments (achievements against targets):

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Grievances received that are addressed within two months of receipt	Percentage	0.00 28-Nov-2017	100.00 18-Dec-2023		98.00 18-Dec-2023

Comments (achievements against targets):

B. KEY OUTPUTS BY COMPONENT

Note: The outcomes listed below derive from the final PDO statement, after the June 2014 restructuring.

Objective/Outcome 1: Increased access to clean energy in rural areas through renewable energy	
Outcome Indicators	**1**. 5.28 million households, farmers, and businesses having access to clean energy services **2**. Additional 85 MW generation capacity of renewable energy (other than hydropower) constructed **3**. 9.56 million direct Project beneficiaries **4**. 71.27 percent female beneficiaries **5.** 5.43 million people provided with access to electricity under the project by household connections **6**. 4.05 million people who gained access to more energy-efficient cooking and/or heating facilities
Intermediate Results Indicators	**1.** 1.2 million solar home systems installed **2.** 9,802 connections made through mini-grid systems and captive plants **3.** 1,303 solar irrigation pumps installed **4.** Collection efficiency of SHS PO at 83% **5.** 4.05 million ICS purchased by households **6.** 10,907 domestic biogas plants installed **7.** SREDA operational with core staff hired **8.** 2,321 female staff in POs hired under the project **9.** 98 percent of grievances received under the project's framework are addressed within two months of receipt
Key Outputs by Component **(linked to the achievement of the Objective/Outcome 1)**	*Component A* 1. New solar home systems installed 2. New connections made through mini-grid schemes 3. New connections made through captive plants

	4. New solar irrigation pumps installed *<u>Component B</u>* 1. New higher efficiency cookstoves purchased by households 2. New biogas plants for cooking installed *<u>Component C</u>* 1. Strengthened renewable energy sector institutional framework 2. SREDA operationalized with key core staff appointed 3. Sectoral studies, guidelines and frameworks developed (*See Annex 6.1. for an exhaustive list)*

ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION

A. TASK TEAM MEMBERS

Name	Role
Preparation	
Zubair K.M. Sadeque	Task Team Leader(s)
Ishtiak Siddique	Procurement Specialist(s)
Mohammed Atikuzzaman	Financial Management Specialist
Shakil Ahmed Ferdausi	Social Specialist
Sabah Moyeen	Social Specialist
Iqbal Ahmed	Social Specialist
Supervision/ICR	
Tanuja Bhattacharjee, Jari Vayrynen	Task Team Leader(s)
Fatema Samdani Roshni	Procurement Specialist(s)
Mohammed Atikuzzaman	Financial Management Specialist
Jichong Wu	Team Member
Shahnun Nima Tania	Procurement Team
Sreyamsa Bairiganjan	Team Member
Vidya Venugopal	Counsel
Amit Jain	Team Member
Md Istiak Sobhan	Environmental Specialist
Cecile Pot	Team Member
Barbara Ungari	Team Member
Joonkyung Seong	Team Member
Shourov Kumar Sharma	Procurement Team
Srivathsan Sridharan	Team Member
Raluca Georgiana Golumbeanu	Team Member

Md. Tafazzal Hossain	Procurement Team
Sabah Moyeen	Social Specialist
Ashok Sarkar	Team Member
Qingtao Yang	Team Member
Zubair K.M. Sadeque	Team Member
Razia Nasreen Sultana	Procurement Team

B. STAFF TIME AND COST

Stage of Project Cycle	Staff Time and Cost	
	No. of staff weeks	US$ (including travel and consultant costs)
Preparation		
FY12	3.000	3,409.20
FY13	6.925	14,398.15
Total	**9.93**	**17,807.35**
Supervision/ICR		
FY13	24.525	73,112.47
FY14	35.150	139,346.12
FY15	20.375	133,131.28
FY16	28.584	79,803.38
FY17	16.487	115,283.77
FY18	27.445	169,409.54
FY19	27.993	362,197.39
FY20	31.505	279,827.39
FY21	38.843	241,378.17
FY22	37.214	162,608.41
FY23	25.969	118,919.33
FY24	37.008	178,112.77
Total	**351.10**	**2,053,130.02**

ANNEX 3. PROJECT COST BY COMPONENT

Components	Amount at Approval (US$M)	Actual at Project Closing (US$M)	Percentage of Approval (US$M)
Access to Electricity	309.2	340.19	85.8
Household Energy	46.3	85.41	12.8
Sector Technical Assistance	5.0	13.50	1.4
Total	**360.5**	**439.10**	**100**

ANNEX 4. EFFICIENCY ANALYSIS

ECONOMIC AND FINANCIAL ANALYSIS

Solar Home Systems Component

Table 1: Summary of SHS Program Performance Indicators

	Appraisal	ICR	Unit
SHS Households Benefitted	550,000	1,204,912	#
SHS Capacity	20, 40, 50, 85	37 different sizes, from 10 to 300	Wp
Project			
Economic NPV	225,031*	42,624**	Millions BDT
Economic IRR	43%*	49%**	%
Household Benefits			
Financial IRR to Households	25%	25%	%
Benefits to Partner Organizations			
NPV @ 10%	2,656	35	Millions BDT
IRR	16%	20%	%
Global Benefits			
CO2 Emissions offset	95,325	169,723	tons CO2/year

*Includes low CO2 Emissions, calculated in 2012 using 5 USD/ton factor and 2.41 tCO2e/kl kerosene emissions saved per liter factor directly applied to counterfactual kerosene consumption
** Does not include Shadow Pricing CO2 Emissions Saved Benefits, calculated using modern World Bank methodology. For Low Shadow Pricing, the NPV and IRR rise to 49,994 million BDT and 187%, respectively. Meanwhile, for High Shadow Pricing, NPV and IRR are 58,575 million BDT and 562%, respectively.

1. **Proposed SHS Investment and Alternative.** The Solar Home System (SHS) comprises of a solar panel of varying sizes from 10 to more than 300 Wp each, with appropriately sized controller and batteries, wiring, and efficient CFL or LED lamps and outlet(s) for supplying power to small appliances such as a radio or TV. The amount of electricity produced is directly proportional to the size of the solar panel. The SHS replaces fuel-based lighting, most often kerosene lighting, and disposable or rechargeable batteries for operating small appliances. The light quality from CFL or LED lamps is far superior to lighting from kerosene lamps, so users gain considerable benefits from superior lighting, especially for reading, general illumination and removing a fire hazard posed by kerosene lamps.

2. **Project Economic and Financial Viability.** The economic and financial analysis is based on the supply and installation of 1,204,912 RERED II-funded solar home systems of varying capacities for a period of over 6 years, beginning January 2013 and ending September 2018.

3. Economic internal rates of return were used to assess the viability of SHS where it displaced kerosene lighting and rechargeable batteries. The economic analysis took into account the economic cost of the SHS, the replacement costs of key components and O&M services. Analyses are done in BDT. The benefits are accrued due to the avoided cost of kerosene for lighting and charging batteries. Kerosene consumption data was based on a survey undertaken by IDCOL as part of the "LIVING IN THE LIGHT: THE BANGLADESH SOLAR HOME SYSTEMS STORY" World Bank study. Kerosene retail and economic prices tracked throughout the years were used.

4. From both an economic and financial viewpoint the project has high and robust internal rates of return (IRR). Even without considering consumer surplus benefits, the economic IRR is 49%. The financial IRR is 46%. Meanwhile, average kerosene household consumption would have to drop to 20 liters per year for the economic IRR to reach 10% (NPV=0). Such low levels of consumptions have not been observed in Bangladesh among households considering purchasing SHS. Finally, these results are an improvement on the appraisal economic IRR of 46% and financial IRR of 26%.

5. **PO Viewpoint.** From the viewpoint of the POs, the SHS business is financially attractive, with Modified Financial IRR for the POs of 20% (assuming a finance rate of 6 percent and reinvestment rate of 15 percent). It assumes IDCOL financing of 60-70% of debt at 6 percent with loan tenor of 6 years with a 2 year grace period. The PO extends loans to the households at 15 percent. The MIRR result is an approvement on the value found at appraisal of 16% (a finance rate of 9 percent and reinvestment rate of 15 percent).

6. **Household viewpoint.** The market response in this demand driven project in itself provides a high degree of confidence that individual households find that the SHS are attractive investments. The 20 Wp SHS offers 6 hours of lighting from two 7 Wp SHS (12 light hours). Larger systems offer 12 to 20 light-hours per day. 50 and above Wp systems also offer additional hours of TV viewing or other equivalent services.

7. A detailed financial analysis for individual systems finds that SHS are financially attractive with positive IRR and highly positive NPV for all systems (Table 2).

Table 2: Financial Viability of Individual SHS

Appraisal	20 Wp	40 Wp	50 Wp	85 Wp
Financial NPV @ 10% (BDT)	13,151	8,728	14,036	13,206
FIRR	47%	22%	25%	20%
ICR	33.8 Wp Average SHS size	150 Wp	300 Wp	
Units	[redacted]	14	1	
Financial NPV @ 10% (BDT)	26,684	59,043	-64,251	
FIRR	25%	48%	3%	

8. The costs to households include the initial cost of supply and installation, O&M and replacement of battery, and controller over time. The benefits to a household comprise of avoided purchase of kerosene lanterns, batteries, kerosene fuel, and battery recharging costs.

RAPSS Component

9. The RAPSS portfolio financed by IDCOL included 12 solar photovoltaic (PV) powered mini-grids, 1,303 solar irrigation pumping, and 3 biomass gasification and biogas power plants.

10. As it was done at appraisal and to facilitate comparison, economic cost effectiveness analyses were completed against diesel or kerosene generation as the alternative providing the same levels of service. Cost effectiveness analysis, rather than EIRR computation was undertaken as the type of service (electricity or water delivery), and the service levels (kWh or cubic meters of water supplied) from the renewable energy options and the diesel/kerosene alternative are identical – thus the benefits are identical.

Solar Mini-grids

11. A solar photovoltaic mini-grid comprises of a large, typically ground-mounted solar PV array, batteries, a back-up diesel generator, and a distribution network connecting customers. An example of such a mini-grid is the 100 kWp Sandwip Island solar PV system to serve up to 400 consumers that include the small shops, school, health center and 4-5 residences surrounding Enam Nahar Market.

12. Average consumption per customer from a solar mini-grid is significantly greater than from a SHS. For example, average consumption per customer in the Sandwip Island scheme currently could be about 1000 kWh per year (when all 400 customers are connected, the average would be up to 400 kWh/year). This level of consumption would be suitable for small enterprises and industry. In contrast a 50 Wp SHS delivers only 80-90 kWh per year suitable for providing basic lighting and electricity services. However, a solar mini-grid, or even a biomass gasifier or diesel mini-grid require certain pre-requisites to be fulfilled – there should be a significant number of customers located in a relatively compact area which can be connected by a grid network at low cost. The electricity demand per customer should be high including daytime loads, which implies significant business or industry demand, rather than only household demand. The mini-grid location should also be far from the grid, if not, it would be more economic to meet the demand through an REB grid extension.

13. At appraisal, it was theorized that since the quality and level of electricity service from a solar mini-grid would be the same as from an appropriately sized diesel mini-grid, an economic cost-effectiveness analysis should be undertaken to verify that electricity from a solar mini-grid is less costly than from a diesel mini-grid providing the same level of service. The financial analysis of the solar mini-grid estimated the financial internal rate of return for the investment and the levelized financial cost of electricity from each mini-grid alternative. However, on the ground reporting done by IDCOL has since indicated that the mini-grids would largely replace kerosene lighting costs, and that diesel and other displaced sources of energy were of negligeable importance. As such, kerosene use was the alternative fuel considered in the ICR analysis.

14. An analysis was conducted for a representative mini-grid based on weighted averages of the twelve mini-grids implemented. It serves 911 customers using 1.00 kWh/day/customer. Required PV system size is 240 kWp with a 133 kW diesel serving as a backup to recharge the batteries for exceptionally cloudy/rainy

periods. Total initial capital costs amount to 92.9 million BDT. A back-up diesel is included to provide greater reliability, supplying 10 percent of the electricity. Life of electronic components, as well as batteries, is 7 years. The counterfactual is kerosene lighting, priced out throughout the years. Details of the financial analysis are below.

Table 3: Mini-grid Financial Analysis

Financials	Weighted Average Representative SMG
Sponsor's Equity	18,577,222
Loan	27,865,832
Grant	46,443,054
Interest Rate	6%
Loan Tenor-Years	10
Principal Grace-Years	2
Projected IRR on equity in Financial Model	15%

15. At appraisal, an analysis was conducted for a representative mini-grid serving 500 customers using 0.77 kWh/day/customer based on expected maximum demand at the Manikgonj project. Required PV system size is 103 kWp with a 5 kW diesel serving as a backup to recharge the batteries for exceptionally cloudy/rainy periods. The alternative diesel generators are 2 x 80 kW assuming peak coincident load per customer of 130 W and diesels operate at 80% of their rated capacity. Two diesel generators are used to ensure adequate availability and reliability comparable to a solar mini-grid. Specific fuel consumption is 0.35 liters/kWh and 2% real fuel cost escalation is assumed. Solar PV installed economic cost of $3.72/Wp inclusive of distribution cost and BDT 0.66/Wp for taxes. Battery accounts for $0.81/Wp (exclusive of taxes). Distribution network adds $0.59/Wp to the cost. A back-up diesel is included to provide greater reliability. It is assumed that the back-up supplies 2.5 percent of the electricity. Life of electronic components, batteries is 5 years and diesel generator is 10 years. For financial analysis a 50% grant and tariff of 32 BDT/kWh is assumed based on Sandwip experience, with tariff escalation equal to half the diesel fuel escalation, plus a fixed monthly charge of BDT 100/customer. The diesel-only generators installed cost is 20,000 BDT per kW plus cost of distribution which is the same as for the solar mini-grid.

16. The analysis demonstrated that operating expenditure of solar PV mini-grids is cheaper than costs spent on kerosene. However, the heavy initial costs for mini-grid construction makes it altogether a significantly more expensive source of energy than kerosene lighting. See results in Table 4 below.

Table 4: Solar PV Mini-grid Economic and Financial Results

		Diesel/Kerosene		Solar		
Criteria	Condition	Appraisal (Diesel)	ICR (Kerosene)	Appraisal	ICR	Units
Levelized economic cost of electricity	Base Case	47.7	10	38.4	51	BDT/kWh
Project FIRR		[redacted]		13.7	1	%
Financial Levelized Cost or electricity		42.7	9.5	30.8	51.1	BDT/kWh
Breakeven PV system cost				4.95		USD/Wp
				33% higher	> 300 % lower	higher/lower

17. At appraisal, the tariff of 32 BDT/kWh was significantly greater than the tariff charged by a BPS, and in the ICR representative solar PV mini-grid, consumers are paying a tariff of 30 BDT/kWh. A significant risk in these projects is that customers may not want to pay such high tariffs if nearby BPS customers are paying a much lower tariff. This risk may limit the demand for solar mini-grids. Willingness to pay assessments are crucial.

Solar PV Irrigation Pumping Sub-Component

18. Solar PV irrigation pumping scheme comprises of a PV array on fixed or tracking supports powering a deep well irrigation pump through a variable frequency inverter. Located centrally among the fields to be irrigated, water is distributed through PVC pipes and open channels to the fields. Owned and operated by a PO, the farmers are charged an annual fee based on the area of land to be irrigated. These PV pumps displace diesel pumps. IDCOL has financed a number of such schemes and the response from POs and from farmers are said to be positive. An example of such a pumping scheme is the 3.5 million BDT, 8.4 kWp solar PV array powering a 7.5 kW submersible pump located in Thamrai Upazila in Village Rehatet, Post Shreapur, Dhaka District. It provides irrigation water to 100 bighas (33 acres) and charges farmers 5,000 BDT/bigha/year.

19. The alternative to a solar irrigation system is a diesel generator powered deep well irrigation pump in the same location delivering water to the same fields. There are millions of diesel irrigation pumps operating in Bangladesh.

20. An economic cost effectiveness analysis verified that solar PV irrigation pumping was lower cost compared to diesel pumping when delivering the minimum quantity of water required by the farmers. As the solar PV array has greater capacity than the requirements of the farmers, given their current irrigation practices, the cost of water, should the full capacity of the solar pumping system be usefully utilized, was also computed. The results are summarized below in Table 4, with actual costs, beginning in 2013, considered until 2024. For subsequent years, conservative projections based upon anticipated long-term inflation were utilized. Over 20 years, the 18.5 kW solar pump can potentially deliver 15,695,000 cubic meters of water for a levelized cost of 0.57 BDT/m^3, while the alternative diesel pump can only provide 1,445,400 cubic meters of water for a levelized cost of 1.23 BDT/m^3. This is an improvement on the analysis done at appraisal, wherein a 5.5 kW solar pump was considered, delivering 3,679,200 cubic meters of water over 20 years for a levelized cost of 0.96 BDT/m^3, better than the considered diesel alternative of the time, delivering 2,943,360 cubic meters of water at the levelized cost of 1.10 BDT/m^3. See further results in Table 5 below.

Table 5: Solar PV Irrigation Pumping Summary Results

	Units	Diesel Irrigation		Solar Irrigation	
		Appraisal	ICR	Appraisal	ICR
Capital cost	Thous. BDT	20	33	2,139	8,687
Quantity of Water Delivered	m3/year	147,168	72,270	147,168	784,750
Cost of Water over 20 years*	BDT/m3	2.61	2.7	1.7	1
		Appraisal	ICR	Appraisal	ICR

Breakeven diesel fuel cost	BDT/liter	49.5	0.5% of Diesel Price	49.5	0.5% of Diesel Price

*To match solar pump production over 20 years, 11 are required diesel pumps are required. At appraisal, the number was assumed to be 10.

21. The following table presents details on typical solar irrigation projects implemented by IDCOL.

Table 6: Solar PV Pumping Financials

Indicators	Acceptable Range
WACC	8.70%
NPV	>0
IRR	11%-13%
Equity IRR	13%-14%
DSCR	Minimum 1.20
Payback Period	<10 years

Biomass Gasification / Bio-Electricity Power

22. Producer gas from a biomass gasifier, after cleaning can be fed into an internal combustion engine to generate electricity. IDCOL has financed three projects to-date.

23. An economic and financial avoided cost analysis was conducted for a representative 400 kW biogas power plant of Dutch Dairy Ltd. The economic analysis demonstrated that the biogas plant is costs effective compared to a diesel and LPG-based power source. The endeavor follows the cost effectiveness analysis was conducted for a representative 200 kW biomass gasifier plant at appraisal, but it does not completely validate it, as though the levelized cost of electricity for the project is lower than that of its alternative, it is not by much, and certainly does not compare to the corresponding gap estimated at appraisal. The significantly higher initial capital costs for the Dutch Dairy plant that the appraisal gasifier's most likely explains the discrepancy. See Table 7 for further details.

24. On a financial basis, the effectiveness of the project is still a mystery, as the data provided by IDCOL for the Dutch Dairy bio-electricity project is only includes a general operating expenditure value, not one broken down so as to understand and analyze the specific fuel costs. At appraisal, the levelized financial electricity cost for the gasifier is 12.0 BDT/kWh compared to 25.34 BDT/kWh from a diesel generator. The breakeven economic cost of diesel, when the electricity cost from diesel is equal to that from the gasifier is 26.4 BDT per liter of diesel (compared to the economic cost of diesel fuel of 82.5 BDT per liter. On a financial basis the breakeven diesel fuel cost is 27.6 compared to financial cost of diesel of 61 BDT per liter.

Table 7: Summary results for Biomass Gasification / Bio-Electricity Power

	Units	Diesel Electricity		Bio-electricity	
		Appraisal	ICR	Appraisal	ICR

Capital cost	Thous. BDT	10,447	16,800	18,105	189,909
Quantity of Electricity Delivered	kWh/year	1,168,000	1,168,000	1,168,000	1,168,000
Financial Cost of Electricity	BDT/kWh	25.34	24.3	12.03	23.83
Breakeven diesel fuel cost		Appraisal	ICR		
Financial	BDT/liter	27.6	92% of the cost of diesel across the years	61	
Breakeven bio fuel cost	BDT/kg | BDT/cylinder	13.3 BDT/kg (biomass)		-	

25. To obtain the attractive returns for biomass gasifier generated power requires the plant to operate reliably over the long term, with access to predictably priced biomass fuel, and to be well managed. Internationally, small biomass gasifier power plants have had a spotty record, therefore careful design, well trained operators/managers and fuel that is dry and properly managed is essential to its reliable operation. Biomass financial fuel cost needs to be four times higher for its electricity cost to equal that of diesel electricity (in India, for example, biomass fuel price increased six-fold over a ten-year period in areas with significant biomass power generation).

26. The biomass gasifier would not be a net emitter of C02. However, as they are mainly expected to use agricultural residues, a sustainable source of biomass fuels, they may not qualify for full carbon credits. Common available biomass gasifiers do require some amount of water consumption for cleaning the gas prior to sending it to the engine-generator. The effluent has to be carefully treated and safely disposed.

Household Energy Component

27. Improved cookstoves and biogas stoves that displace traditional stoves save considerable amount of biomass cooking fuels. Importantly, they have very significant environmental and health benefits, especially for women and children. Improved cookstoves and biogas stoves result in significant reduction of indoor air pollutants such as small particulates, and toxic pollutants. The WHO estimated that as much as 3.6 percent of the total burden of disease in Bangladesh is attributable to exposure to indoor air pollution; 32,000 children below 5 years of age die annually due to acute lower respiratory infections, and 14,000 adults die due to chronic obstructive pulmonary disease.

28. An IDE survey conducted for IDCOL identified very significant benefits for biogas plants in terms of improved health, socio-economic status, reduced workload for women, and enhanced agriculture and environment. The users have reported significant health benefits resulting from reduced air pollution and the associated eye and respiratory infections. One notable benefit is the reduction of fire-induced accidents resulting from non-use of firewood and other traditional fuels. There were significant benefits from time savings.

Improved Cook Stoves

29. The analysis is conducted at the national and household level for a program that would support the replacement of traditional wood burning stoves with improved cook stoves (ICS). The economic analysis uses project TA costs, household investment and maintenance as costs, and the value of fuel savings and health expenditure savings as benefits. The financial analysis from the household perspective considers only the direct investment and maintenance cost to the household and fuel savings. The ICS program has positive economic and financial benefits as shown below in Table 8.

Table 8 Economic and Financial Analysis Results for Improved Cook Stoves

	Project (Economic)		**Households (Financial)**	
Millions BDT	**Appraisal**	**ICR**	**Appraisal**	**ICR**
Total Costs	2,319	8,415	2,140	4,467
Total Net Benefits	740	60,940	1,459	60,085
NPV	419*	25,425.77**	820*	25,355.78**
EIRR and MIRR	36.40%*	> 1000%**	39.70%*	441%**
Project Life	7	15	7	15

* Includes CO2 Benefits, calculated through a 0.66 carbon emissions saved per stove per year, times 5 USD/tCO2e.
** Does not included CO2 emissions saved benefits, which, under Low Shadow Pricing, lead to an NPV of 40,087 million BDT and almost 10000% IRR, while High Shadow Pricing leads to an NPV of 54,657 million BDT and over 15000% IRR.

30. The economic and financial results are robust though sensitive to avoided fuel costs. These returns may even feel overwhelming and are substantially greater than what was estimated at appraisal. However, it was already then stated that the assumed 574 BDT yearly biomass fuel cost undervalued true, on-the-ground expenditure. Recent 2022 survey data suggest an approximate cost of 3,138 BDT. Furthermore, the ICR ICS analysis operated on a 15-year project life, not the 7 at appraisal, taking into account the 11 years ICS were deployed, as well as their 5-year lifespan. Finally, at appraisal, rural biomass expenses were found to have to decline to BDT 2.43/kg (compared to assumed cost of BDT 3/kg) for the financial net present value (at 10%) to be zero. The ICR analysis found that yearly biomass fuel would have to decrease to 93% of the aforementioned 2022 cost in order for net present value to reach zero.

Biogas Plant with Biogas Stove

31. The analysis is conducted at the national and household level for a program that supported the replacement of traditional wood burning stoves with 10,907 (estimated 20,000 at appraisal) biogas plants and biogas stoves. The analysis is based on households acquiring a biogas plant with a daily gas production of 2.8 m3 required to cook three meals a day. It would require daily feeding rate of 60 kg of dung per day. 32 Households with 5 or more head of cattle would have sufficient dung to support a gasifier producing 2.8 m3/day of biogas.

32. The economic analysis uses project capital, TA investments and household investment as costs and the value

of fuel savings, along with value of nutrients (displacing purchased fertilizer), domestic labor savings, value of C02 emission reductions, and health expenditure savings due to reduced indoor air pollution as benefits. The biogas displaces purchased fuelwood burned in a traditional cook stove. The financial analysis done from a household perspective uses only the direct cost to households as costs, and the fuel savings as benefits. The project is economically and financially viable. For the financial analysis, Project Investment costs (project capital and TA investments), were significantly more costly than anticipated at appraisal. So much so that at closure, these costs overwhelm the combined loss of economic health and domestic labor benefits, which are similarly not included in the financial analysis. Finally, the ICR Biogas analysis has a life of 25 years as opposed to 15 at appraisal, taking into full consideration both the 15-year life of small biogas plants and the 11-year period during which they were deployed by the project. The results are summarized in Table 9.

Table 9: Biogas Economic and Financial Results

	Project (Economic)		**Households (Financial)**	
Millions BDT	**Appraisal**	**ICR**	**Appraisal**	**ICR**
Total Costs	265	1,431	540	727
Total Net Benefits	1,254	4,964	1,519	3,254
NPV	356*	1,014.47**	535*	766.45**
EIRR and MIRR	27.30%*	27%**	15.10%*	50%**
Project Life	15	25	15	25

* Includes CO2 Benefits, calculated through a 0.66 carbon emissions saved per plant per year, times 5 USD/tCO2e. EIRR drops to 26% without them.

** Does not included CO2 emissions saved benefits, which, under Low Shadow Pricing, lead to an NPV of 1,228 million BDT and almost 29% EIRR, while High Shadow Pricing leads to an NPV of 1,340 million BDT and over 30% EIRR.

ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

ANNEX 6. SUPPORTING DOCUMENTS (IF ANY)

ANNEX 6.1. List of knowledge products and reports related to the RERED II Project

Recognitions provided to the project

Financial Innovation Awards in the category of Best Sustainable Finance Initiative for the IDCOL Solar Irrigation Pump Project – London Institute of Banking and Finance (2019)

Karlsruhe Sustainable Finance Awards in the category of "Outstanding Sustainable Project Financing" for the Solar Irrigation Pump Projects – European Organization for Sustainable Development (EOSD, 2020)

Asian Power Awards in the category of "Solar Power Project of the Year – Bangladesh" for the Solar Irrigation Pump Projects – Asian Power Magazine (2020)

Knowledge products describing the achievements and lessons learned of the RERED II

Anil Cabraal, William A. Ward, V. Susan Bogach, Amit, Jain (2021), *Living in the light: the Bangladesh Solar Home Systems Story,* World Bank Study

Berkeley Air Monitoring Group (BAMG) in partnership with International Centre for Diarrheal Disease Research, Bangladesh (ICDDR, B), *Indoor Air Pollution (IAP) study*

Mitra, A.; Alam, M. F.; Yashodha, Y. (2021) *Solar irrigation in Bangladesh: a situation analysis* Report, International Water Management Institute (IWMI)

Monzur Hossain, Azreen Karim, *Impact Assessment of Solar Irrigation Program of IDCOL* (2019), Bangladesh Institute of Development Studies

Buisson, M.-C.; Mitra, A.; Osmani, Z.; Habib, A.; Mukherji, A. (2023) *Impact assessment of Solar Irrigation Pumps (SIPs) in Bangladesh: a baseline technical report,* International Water Management Institute (IWMI)

Wakil Ahmed Arnob, Shisher Shrestha, Md. Abdullah Al Matin, Archisman Mitra, *Grid Integration Report – Bangladesh (*2023*),* IMWI, IDCOL, Solar Irrigation for Agricultural Resilience, Swiss Agency for Development and Cooperation

Beyond Diesel: Navigating an Equitable and Sustainable Irrigation Landscape in Bangladesh (2024) – Report on a National stakeholder workshop, IWMI, IDCOL, NGO Forum for Public Health

Knowledge products enabled[36] by the RERED II under the Technical Assistance Component
(Source – Power Cell)

Sl No.	Package Description	The Consultant	Short note of the study outcomes
01.	S-04: Technical Audit Study for old Power Plants of Bangladesh	Steag Energy Services (India) Pvt. Ltd. In association with (Sub consultant) Development Design Consultants Limited (DDC) (Bangladesh)	Low efficient 16 Power Plants under BPDB were set the KPIs in respect of Heat Rate Values and consumption factor. Overall, performance improvement addressed of the power plants through technical auditing.
02.	S-05: Individual Consultant to Assist Power Cell (EIA Consultant)	Dr. Md. Nazim Uddin	The environmental consultant carried out environmental screening/assessment along with analysis of alternatives, prepared the Environmental Management Plan (EMP) and contributed to bidding document for incorporation of EMP cost for Rural Electricity Transmission and Distribution (T&D) Project, financing by the World Bank. In addition the consultant trained up the relevant staff of REB, Palli Bidyut Samities (PBSs) and PGCB in environment management through Environment and Social Management Framework (ESMF) for transmission and distribution project implementation.
03.	S-07: Cumulative Environmental Impact Assessment (CEIA) at Siddhirganj Power Hub	ERM INDIA PRIVATE LIMITED in association with (Sub consultant) EQMS Consulting Limited	VEC (Air Quality, Ground and Surface Water Quality, Land (Soil & Sediment Quality and its Use), Ecology & Biodiversity, Socioeconomics, Community Health) were assessed in the Siddhirganj & Meghnaghat Industrial area for findings the Power Plants pollution rate. The areas were more polluted by other industries than power plants.
04.	S-10: CFL Bid process Consultant	Dr. Gert Wemmer	Efficient Lighting Initiatives for Bangladesh (ELIB) program under which 10 million Compact Fluorescent Lamps (CFLs) were distributed in high-energy demand areas of the country in exchange of existing incandescent lamps being used in the residential homes. In this successful initiative of the government, the right quality of CFLs conforming to the technical specifications and performance test standards were supplied under this program by Bangladesh Rural Electrification Board (BREB).
05.	S-18: Digitalization of Sustainable and Renewable Energy Development Authority (SREDA)	Nano Information Technology	SREDA's institutional and capacity building development through digitalized and computerized platform such as Human Resource Management *Systems* (HRMS) includes *leave* management, training

[36] Support was provided directly and indirectly (reimbursement of consultancy services and/or technical inputs).

			management, vehicle management, performance evaluation *management* etc., database for renewable energy source identification, R&D, fund collection, incorporation of private sectors, legal framework, promotion, achievement etc., identification of scope of energy efficiency using promotional actions (Public awareness, using energy efficiency devices etc) or by regulation, legal framework etc., Accounting & Financial Management System (Accounting, Budget, Loan, Revenue & Development fund, Provident fund etc.) and Asset Management System. Also developed and upgradation and host an interactive and dynamic web site (both Bangla & English) and provided webmail service for the employees.
06.	S-19.1 & S-20.1: Accounting & Financial Consultant for BERC S-19.2: Consultancy Support for BERC (Gas) S-19.3: Consultancy Support for BERC (Petroleum) S-19.4: Consultancy Support for BERC (License application review, energy audit and TQS) S-19.5 & S-20.2: Consultancy Support for BERC (Legal) S-19.6 & S-20.3: Consultancy Support for BERC (Translation) S-19.7: Consultancy Support for BERC (IT System, Database & Website Management)	1. Mr. Md. Khalilur Rahman 2. Mr. Manjur Morshed Talukder 3. Mr. AHMA Majid Rahimi 4. Mr. Hossain Shaheen Meer Akram 5. Mr. Md. Asaduzzaman and Mr. Md. Maznul Ahan 6. Mr. Muhammad Abdul Mannan and Mr. Md. Rashel Siddique 7. Mr. Md. Wali Ahad	Under the institutional and expertise development of Bangladesh Energy Regulatory Commission (BERC) in the field of commission governance, management, budgeting, regulatory monitoring, accounting, tariff setting, asset valuation, public hearing and strategic planning and preparation and upgrading of regulations, codes, standards, regulatory concept papers, energy fund documentation, manuals and procedures to facilitate BERC's day-to-day activities.
07.	S-22: Feasibility Study construction of Aminbazar-Maowa-Mongla Transmission Line & Associated Substation and Anwara-Meghnaghat Transmission line & Associated Substation	PGCIL, India	PGCB prepared the DPP based on the FS and implemented the project accordingly.

08.	S-23: Environmentally Safe Disposal of CFL	Ernst & Young, India	CFLs contain toxic mercury as well as other substances like glass and other metal parts. When broken, compacted, crushed, or disposed o ff improperly, fluorescent bulbs may release mercury into the air, water, and land, posing significant threats to people and the environment. The proper disposal guideline formulated for the CFLs disposal hazard mitigation. Department of Environment (DOE) is the key implementation authority in Bangladesh for any environmental hazard measures and its mitigation. The guidelines helped DOE for action taken against the disposal of CFL and proper management in this regard accordingly.
09.	S-27: Master Plan for the Ghorasal, Goalpara and Baghabari Generatrion Hub	Engineers and Consultants Bangladesh Ltd. (ecbl)	Optimization of the Land and properties of the generation hubs used and planned for future and implementation accordingly.
10.	S-32.1: Individual Consultants Civil Engineering for EGCB S-32.2: Individual Consultants Electrical Engineering for EGCB S-32.3: Individual Consultants Mechanical Engineering for EGCB S-32.4: Individual Consultants I & C Engineering for EGCB S-32.5: Individual Consultants Geo Technical for EGCB	1. Mr. James G. Smith 2. Mr. Worth Edwards 3. Mr. Martin Tormly 4. Mr. Ronald F. Tipton 5. Mr. Fernando Pons	Design & drawing approval, engineering, project management and construction oversight for Electricity Power Generation Company of Bangladesh (EGCB)'s Siddhirganj 335 MW Combined Cycle Power Plant along with the associated auxiliaries and ancillary equipment to complete the project.
11.	S-33: International Procurement Consultant for CFL	Mr. Narayan D. Sharma	Strengthen the procurement process by including an international procurement expert in the bidding process for Efficient Lighting Initiatives for Bangladesh (ELIB) under BREB.
12.	S-34: ESMF for Grid and Distribution System	Bureau of Research, Testing and Consultation (BRTC) under Bangladesh University of Engineering and Technology (BUET)	The guidelines carried out about environmental and social screening, analysis of alternatives, and detail Environmental and Social Impact Assessment (ESIA), including preparing of Environmental Management Plans, Resettlement Action Plans and Tribal Peoples Plans, of subprojects to be implemented under the proposed "Rural Electricity Transmission and Distribution (T&D) project. The T&D project was the main beneficiary of the guideline for smooth implementation and project execution in rural people/area.

13.	S-37: Local Procurement Consultants to assist Power Cell	Mr. Md. Sarifuzzaman	The World Bank Procurement practices and STEP procedures were implemented for the Power Cell project.
14.	S-38: Professional Editor	Mrs. Shireen Kamal Sayeed	The documents related to power sector act, policy, regulation and others were reviewed and improved by the consultant.
15.	S-39: Consultant for Hydro power resource Mapping	Stream Tech Inc., USA	Explored the Hydro Power Potentiality in Hill area, Bangladesh by this study. 6 sites along Sangu River had the peak hydropower generation potential exceeding 10 MW, but the predicted period of plant operation ranges from only 7 to 10 weeks per year (13.3% to 18.3% of the time) for the run-of-river type plant. Increasing the period of operation results in a reduction of total energy output over a year.
16.	S-48: Feasibility Study on 60MW Solar Park at Rangunia, Chittagong	Atlanta Enterprise Ltd. (Bangladesh)	This feasibility study prepared for preparation of DPP and BPDB has taken a Solar Project at the location.
17.	S-51: International Procurement Consultant for new project of REB	Mr. John Richardson	Up-gradation of Rural Electric Distribution System (UREDS) project of BREB under The World Bank umbrella project of "Rural Electricity Transmission and Distribution Project" was executed and implemented by the consultant's inputs.
18.	S-52: International Technical Consultant for new project of REB	Mr. Chrissantha Ratnayake	
19.	S-53: Financial Management Specialist for New project of REB	Mr. Md. Abul Quashem	
20.	S-54: Development & Implementation Support for Bangladesh Efficient Lighting Transformation Program (LED Lighting Program)	Deloitte Touche Tohmatsu India Pvt. Ltd.	Energy efficiency improvement and conservation mechanism, GoB has taken initiatives for exchanging the incandescent lamps (Ils) to CFLs under Efficient Lighting Initiatives of Bangladesh (ELIB) supported by the World Bank. The survey and market assessment found the success of the initiative.
21.	S-59: ICT Road Map for Bangladesh Power Sector	Deloitte Touche Tohmatsu India Pvt. Ltd.	Based on the road map, ICT strategy has been developed for Bangladesh power sector. Enterprise Resource Planning (ERP) was introduced in the power sector and presently, ERP implementation is under way by the GoB fund.
22.	S-60: Survey of Customer Satisfaction under DPDC, WZPDC and one PBS	Micro Industries Development Assistance and Services (MIDAS) (Bangladesh)	Among 5000 customers of residential, commercial, Irrigation class under DPDC, WZPDC and BREB (5000 each) were surveyed in Dhaka, Narayangonj, Gazipur, Manikgonj, Khulna, Jessore and Barisal districts. The

			modernize DISCOs networks will provide better services to the consumers.
23.	S-64: International Individual Environment Consultant for Siddhirganj Power Hub	Mr. Juin D. Quintero	Global standards of environmental factors due to Power Plant implementation were addressed and formulated the practices.
24.	S-66: International Individual Environment Consultant for Repowering of Ghorasal Power Plants (Unit#4)	Mr. Prasad Madhav Modak	Rehabilitation and repowering mechanism of old power plant were standardized considering the environmental factors.
25.	S-67: Consultancy Service for Social Assessment Surveys, Preparation and Implementation of RAPs for proposed ECGSTLP under T &D Project	Development Research Initiative (DRI) (Bangladesh)	Focus Group Discussion (FDG) and Resettlement Action Plan (RAP) prepared for compensation of the affected people and structures. Afterward Enhancement of Capacity of Grid Substations and Transmission Lines for Rural Electrification (ECGSTLP), financing of the World Bank was implemented.
26.	S-68: Feasibility Study on Solar Irrigation System in Bangladesh	Environmental Quality and Management System (EQMS) Consulting Limited (Bangladesh)	The government initiatives for reducing financial burdens, environmental and social impacts for diesel based irrigation pump/system was addressed in the report. Irrigation by using solar system to replace diesel powered irrigation was achieved successfully.
27.	S-69: Reliability Study of Bangladesh Power Grid System	Monenco Iran Consulting Engineers in association with (Sub consultant) Megatech GNBD Dhaka, Bangladesh	National Grid network strengthening and planning in respect of technologies adaptation, skills development and existing system upgradation i.e. grid code.
28.	S-70: Energy Efficiency & Conservation Expert for SREDA	Dr. Gopal Chandra Datta Roy	SREDA’s energy efficiency and conservation mechanism was promoted and sectoral action was taken in this regard.
29.	S-72: Residual Life Assessment (RLA) of Steam Turbine and Generator of Unit#4 Ghorasal Power Plant	KEMA Netherland B.V., Netherlands	Repowered and increased power generation capacity of the units.
30.	S-73.1: International Procurement Expert for Independent Procurement Panel under Ghorashal Unit#4 Repowering Project S-73.2: Technical Expert (Electrical) for Independent Procurement Panel	1. Mr. George Jadoun 2. Mr. Reinaldo “Rene” G. MENDONCA 3. Mr. Rishi Kumar Jain 4. Mr. Bruce K Paley 5. Mr. John Richardson	The World Bank funded “Ghorashal Unit#4 Repowering Project” was implemented.

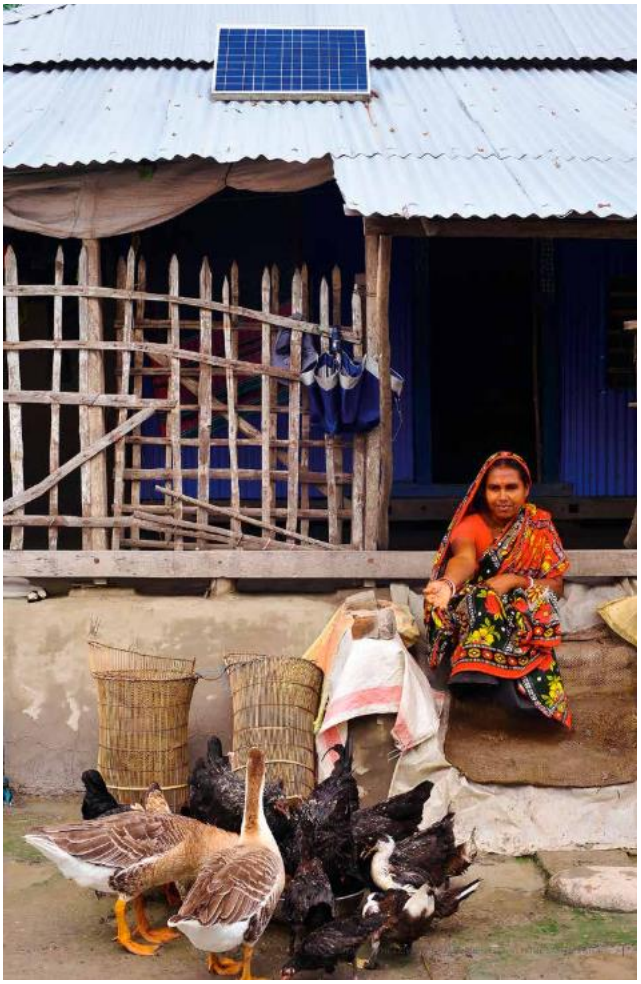
	under Ghorashal Unit#4 Repowering Project S-73.3: Technical Expert (Mechanical) for Independent Procurement Panel under Ghorashal Unit#4 Repowering Project S-73.4: Technical Expert (Boiler & Steam Turbine) for Independent Procurement Panel under Ghorashal Unit#4 Repowering Project S-80: International Procurement Expert for Evaluation Committee to Select Owner's Engineer of Ghorashal Unit#4 Repowering Project		
31.	S-75: Update of Gas Sector Master Plan	Ramboll Denmark A./S in association with (Sub consultant) EQMS Consulting Limited (Bangladesh)	Energy Division inputs under this study for upgradation of the GSM Plan.
32.	S-76 and S-96: International LNG Expert/Consultant for Construction of FSRU Project, LNG Project Team (LNG Cell), Petrobangla	Mr. Oystein Bruno Larsen And Mr. Holger's Kelle	Technical and legal issues were taken for establishment of FSRU project under Energy Division, Ministry of Power, Energy and Mineral Resources of Bangladesh.
33.	S-80: International LNG Legal Expert for Construction of FSRU Project, LNG Project Team (LNG Cell) of Petrobangla	Mr. Tom West, Jr.	
34.	S-84: BPMI Consultant	Mr. Tapos Kumar Roy	Establishment of Bangladesh Power Management Institute (BPMI).
35.	S-85: Consulting Services on Upgradation of Technical and Fiscal Terms of the Model Production Sharing	JV of Resource Development Ltd. (RDL), New Zealand and Development Technical Consultants Pvt. Ltd. (DTCL), Bangladesh	Energy Division inputs under this study for upgradation of Model PSC.

	Contract (PSC) for Oil & Gas Exploration in Bangladesh		
36.	S-86: Feasibility Study of Enhancement & Strengthening of Power Network in Eastern Region of PGCB	PGCIL, India	The FSR focused for preparation of DPP of the "Enhancement & Strengthening of Power Network in Eastern Region of Power Grid Company of Bangladesh (PGCB)" Project.
37.	S-87 & 100: Baseline Survey for Global Multi-Tier-Measurement for Access to Energy	MIDAS, Bangladesh	Reliability of the electricity service (frequency of outages) has been improved, for considering the most households in Bangladesh.
38.	S-88: Residual Life Assessment (RLA) of Steam Turbine and Generator of Unit#6 Ghorasal Power Plant	Steag Energy Services (India) Pvt. Ltd.	Repowered and increased power generation capacity of the units.
39.	S-89: Feasibility Study for development of utility scale Solar PV & Wind Project of EGCB	M/S WinDForce Management Services Pvt. Ltd., India inassociation with Suntrace GmbH (Germany) and EQMS Consulting Limited (Bangladesh)	EGCB has been implementing the Solar project in Sonagazi, District Feni for utilization of the World Bank financing.
40.	S-90: Individual Consultant for Monitoring the implementation of computerized Integrated Financial Management System (IFMIS) of BREB & PBSs	Md. Faruque Sikder	Introduced digital financial management system for BREB.
41.	S-93: Advisory Services for Implementation of Enterprise Resource Planning in Power Sector of Bangladesh	Mr. Rakesh Asthana	Initiatives for introduction of ERP in Power Sector.
42.	S-98: Consulting Services for Assessment of compensation and Preparation of Resettlement Action Plan (RAP) for Transmission Lines in ECGSTLP for T&D Project of PGCB	Development Research Initiative (DRI), Bangladesh	PGCB worked on the RAPs for compensation to the people under the ECGSTLP project implementation.
43.	S-102: Consulting Services for Assessment of	Development Research Initiative (DRI), Bangladesh	BREB worked on the RAPs for compensation to the people under the UREDS; DCSD project implementation.

	Compensation and Preparation of Resettlement Action Plan (RAP) for Design, Supply, Installation, Testing & Commissioning of 33 kV River Crossings under Up-gradation of Rural Electricity Distribution System; Dhaka, Chittagong & Sylhet Division (UREDS; DCSD) Project of BREB		
44.	S-104: Feasibility Study for Electrification of Rangabali Upozilla through Renewable Energy	Infrastructure Development Company Limited (IDCOL) (Bangladesh) in association with Meghraj Capital Advisors Private Limited (MCAPL) (India)	Electrification of island people by solar project implementation.
45.	S-106: Appointment of Consultant for Human Resource (HR) Assessment & Institutional Capacity Development in Power Sector	KPMG, Singapore	Capacity building and skill development was notified for booming power sector in Bangladesh.
46.	S-107: Consulting Firm for Preparation of Resettlement Action Plan (RAP) under Enhancement and Strengthening of Power Transmission Network in Eastern Region Project of PGCB	Knowledge Management Consultants (KMC) Ltd. Bangladesh	RAP implemented by PGCB
47.	S-110: Appointment of Individual Consultant for Preparation of Net Metering Guidelines	Mr. Shariar Ahmed Chowdhury	Integration of Renewable Energy (RE) into the grid has been emphasized for increasing future contribution of RE in the grid.
48.	S-111: Appointment of International Consultant for Implementation of Independent System Operator (ISO)	Monenco Iran Consulting Engineers (Iran)	For ensuring reliable and quality power supply ISO implementation is key factor for a country power system. The Bangladesh ISO model has been development in this study to meet the Government actions as per Electricity Act, 2018.
49.	S-112: Individual Consultant for case study of construction delay of 335 MW	Mr. Sayedul H Choudhury	The case study conducted for investigation of the delays of the construction of power plant. It was found that initial project preparatory stage was more for arranging finance and also EPC

	Siddirganj Power Project		was delayed for implementation works as well.
50.	S-114: Rationalize and update the Power Purchase Agreements (PPA) and Implementation Agreements (IA) of IPPs	Deloitte Touche Tohmatsu India Pvt. Ltd.	The upgradation made as per present requirements and effective manner.
51.	S-115: Appointment of a firm for Central Complain and Feedback Management System	M/S MCC Ltd., Bangladesh	Introduction of customer complain and feedback system through digital platform under Power Division.
52.	S-116: Appointment of Consulting firm for Assessment of Barriers to Implement RE Project and Standardized based Tender & Contract Documents	Atlanta Enterprise Ltd. (Bangladesh)	Solar projects were being implemented faster.
53.	S-119: Update of Resettlement Action Plan (RAP) for Enhancement and Strengthening of Power Transmission Network in Eastern Region Project at Laksam and Muradnagar Under Power Grid Company of Bangladesh (PGCB) Ltd.	KMC, Bangladesh	RAP implemented by PGCB

ANNEX 6.2. Photographs of a sample of the energy solutions disseminated under the RERED II
(Source: ICR Mission, unless indicated otherwise)



Photograph 1. Component A.i. Solar Home System, Final residential beneficiary

Source: Anil Cabraal, William A. Ward, V. Susan Bogach, Amit, Jain (2021), *Living in the light: the Bangladesh Solar Home Systems Story,* World Bank Study

Photograph 2 – Component A.ii. Solar Irrigation Pump Project – Thakurgaon Sadar, Thakurgaon

Photograph 3 – Component A.ii. Solar panels of the Solar Irrigation Pump Project – Thakurgaon Sadar, Thakurgaon

Photograph 4 – Component A.ii. Solar Irrigation Pump Project – Thakurgaon Sadar, Thakurgaon

Photograph 5 – Component A.ii. Biogas/Bioelectricity captive power plant (Kazi Tea Estate) – Tentulia, Panchagarh

This photograph shows one of the components of the biogas/bioelectricity captive plant of the Kazi Tea Estate, and more specifically focuses on the biogas plant (center) using cow dung directly collected from one of the 9 cow sheds of the estate (left).

Photograph 6 – Component A.ii. Biogas/Bioelectricity captive power plant (Kazi Tea Estate) – Tentulia, Panchagarh

This photograph shows the biogas generator room of the project. This gas is used directly by the factory for both its personal use (staff consumption, 8 domestic stoves) and its commercial use (4 commercial stoves involved in sweet making using estate's products). Next to it, a transformer room generates captive electricity based on the biogas generated by the plant. This bioelectricity is used by the estate for its own consumption, complemented by grid electricity when available (e.g. strategy to address load shedding).

Photograph 7 – Component A.ii. Biogas/Bioelectricity captive power plant (Kazi Tea Estate) – Tentulia, Panchagarh

This photograph shows the machinery involved in the biogas generator room of the project. This machinery involves the same anaerobic digestion principle as the one used under Component B. Household Energy (domestic biogas digester), at a larger commercial scale.

Photograph 8 – Component A.ii. Commercial cookstove fueled by the gas directly produced by the biogas/Bioelectricity captive power plant (Kazi Tea Estate) – Tentulia, Panchagarh

Photograph 9 – Component B. Components of a portable Improved Cookstove (portable ICS, on the right, used by beneficiaries who do not own a kitchen room) – Thakurgaon Sadar, Thakurgaon

This photograph shows the technological improvements of the portable improved cookstove, including the insulating design which increases energy efficiency (compared to the traditional stoves used), fume evacuation design and serial number to benefit from after-sale services.

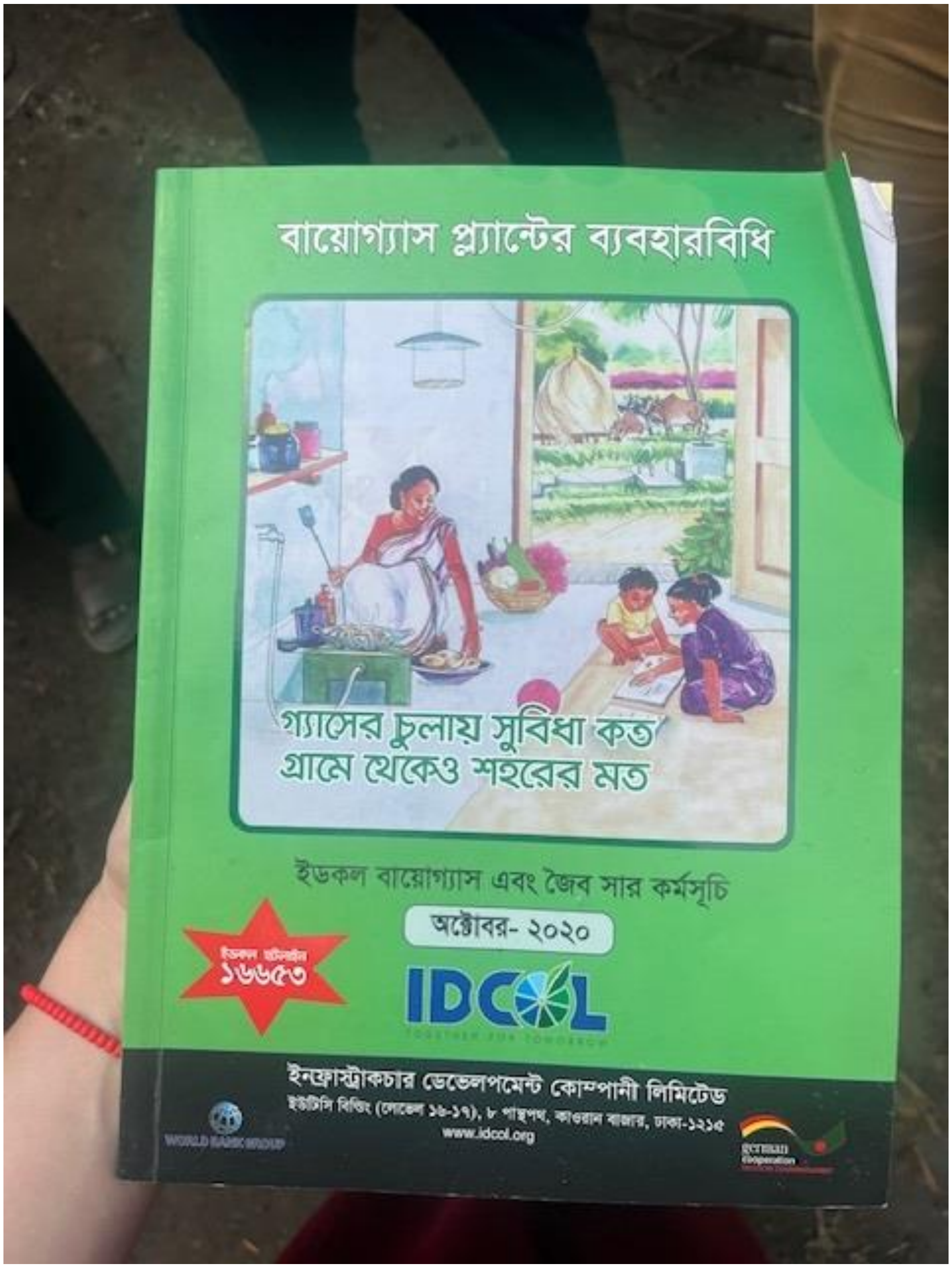
Photograph 10 – Component B. Fixed Improved Cookstove (ICS, used by beneficiaries who own a kitchen room).– Thakurgaon Sadar, Thakurgaon

This photograph shows the technological improvements of the fixed improved cookstove, including the chimney installed to evacuate the fume outside of the kitchen room, the insulating design which increases energy efficiency, and serial number to benefit from after-sale services.

Photograph 11 – Component B. Domestic Biogas Digester – Birganj, Dinajpur (Beneficiary Household's home)

This photograph shows the improved model of the domestic biogas digester installed under the restructured version of this sub-project. This improved design is prefabricated using modern material (in comparison to the former brick-model which required lengthy and unreliable masonry services). The cow dung used in this digestor generates both clean methane used for cooking (evacuated through the pipeline in the center of the photograph, connected to the house) and fertilizer used in the beneficiary's owned field (pipeline on the bottom left corner).

Photograph 12 – Component B. Methane-powered cookstove using Domestic Biogas Digester (right) compared to the biomass traditional stove used before project intervention (left) – Birganj, Dinajpur (Beneficiary Household's home)



Photograph 13 – Component B. Domestic Biogas Digester – Birganj, Dinajpur (Beneficiary Household's home)

This photograph shows the user-friendly manual provided to final beneficiaries to both explain the use of the biogas digester, its benefits, and provide key information on after-sale services (such as IDCOL's call center number, in red). Similar customer documents (including of legal nature) were distributed under other sub-projects to ensure quality after-sale services.

Annex 7 - ADDITIONAL INFORMATION

Additional information on changes to the RERED II project over its implementation

- **In January 2014** – A first indirect restructuring did not target the RERED II project, but targeted two child projects that served to disburse grant financing to Component A (GPOBA Rural Electrification and Renewable Energy – SHS – P119549; and GPOBA Rural Electrification and Renewable Project – P119547 – Minigrids). This restructuring allowed the Project to reallocate funding between disbursement categories to further support the SHS and RAPSS activities, and to include micro-grid as an eligible technology for support under the Project.

  Rationale: The January 2014 restructuring was one of underlying projects under RERED II and did not significantly affect its design.

- **In August 2016,** the Project adjusted elements related to the Trust funds financing the capital buy-down grants needed under component A and component B : such as (i) under TF015034 – extension of the closing date from August 31, 2016 to December 31, 2018; and inclusion of additional funding received from USAID to support this activity; (ii) under the child projects abovementioned: inclusion of "advances" as an eligible disbursement method. This restructuring also allowed to decrease the target for DBD dissemination (33,000 to 16,000).

  Rationale: The changes to the TFs (TF015034, TF01956, TF019157) financing component A and B were introduced to accommodate for additional grant financing received from USAID (extension of closing date of TF015034), and to facilitate the implementation of SHS and RAPSS activities (introduction of advances as a disbursement method). Component B's targets for biogas plants to be installed were revised downwards, due to an increase in the cost of this activity, which mechanically reduced the number of domestic biogas digesters possible to install. Indeed, as quality challenges were identified on the first DBD models initially targeted, alternative and more expensive models were preferred, requiring a higher subsidization.

- **In June 2017,** the following changes were introduced: i) extension of the closing date of the Grant Agreement (TF019156 and TF019157) from June 30, 2017, to June 30, 2018; ii) reallocation of funds to support additional technical assistance to ensure the project sustainability.

  Rationale: The changes suggested under this restructuring aimed at accommodating the emerging financing needs of component A and B.

- **The last restructuring in December 2023** took place in the context of the project closing and had two purposes: (i) cancellation of the remaining IDA-credit funding which was not spent by the closing date; and (ii) reallocation of funds between disbursement categories under the Project.

  Rationale: This last restructuring was mainly aimed at surrendering the undisbursed funds under the project by its closing date.